UNIT – I

Simple Mechanisms

Subject : Theory of Machines (23A03403)

Syllabus: Classification of mechanisms, basic kinematic concepts & definitions, degree of freedom & mobility, Grashof's law, inversions of four-bar and slider-crank chains, limit positions, mechanical advantage, transmission angle, quick-return mechanisms, straight-line mechanisms, universal joint, rocker mechanisms.

1. Basic Definitions

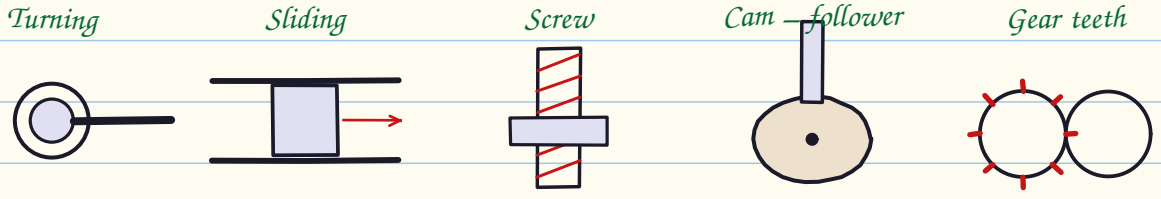
Theory of Machines = study of the relative motion between parts of a machine and the forces acting on them. Divisions : Kinematics (motion without forces), Dynamics (kinetics – forces & motion), Statics.

- **Machine** : combination of resistant bodies that transmits motion and does useful work (engine, lathe). **Mechanism** : a kinematic chain with one link fixed whose purpose is to transmit/transform motion (no concern for energy) – e.g. a clock, typewriter. **Structure** : assembly of resistant bodies with no relative motion.
- **Link (element)** : a resistant body, rigid (assumed) – connecting rod, or flexible (belt, chain) or fluid (hydraulic press). Binary link (2 joints), ternary (3), quaternary (4).
- **Kinematic pair** : two links in contact such that relative motion is constrained.

Classification of kinematic pairs

Basis	Types	Example
Nature of contact	Lower pair (surface contact) / Higher pair (point or line contact)	turning, sliding, screw / cam-follower, gears
Relative motion	Sliding, turning (revolute), rolling, screw (helical), spherical	piston-cylinder, shaft in bearing, ball bearing, nut-bolt, ball & socket

Basis	Types	Example
Closure	Self-closed (geometric) / force-closed (spring, gravity)	journal bearing / cam & follower with spring
Constraint	Completely, incompletely, successfully constrained	shaft in bearing / shaft in foot-step / piston with gas force



lower pair	lower pair	lower pair	higher pair	higher pair
$F = 1$	$F = 1$	$F = 1$	$F = 2$	$F = 2$

- ***Kinematic chain*** : assembly of links and pairs forming a closed loop with relative motion that is constrained. ***Inversion*** : obtaining different mechanisms from the same chain by fixing different links.
- ***Degree of freedom (DOF / mobility)*** : number of independent parameters (inputs) needed to specify the configuration of the mechanism. A rigid body in a plane has 3 DOF ; in space 6.

Classification of mechanisms

- By motion : plane (planar), spherical, spatial. By number of links : 4-bar, 6-bar, 8-bar. By function : function generation, path generation, motion generation. By DOF : single-DOF (constrained), multi-DOF (robot arm).
- By pair type : lower-pair mechanisms (linkages) and higher-pair (cam, gear, ratchet).

2. Degree of Freedom & Mobility

Grübler's (Kutzbach) criterion for plane mechanisms

$$F = 3\,(n - 1) - 2\,j_1 - j_2$$

n = total number of links (incl. frame), j_1 = number of lower pairs (1-DOF), j_2 = number of higher pairs (2-DOF). (For only lower pairs $F = 3(n - 1) - 2j$ — Grübler.)

- $F > 0$: mechanism ; $F = 1$: constrained (single input) ; $F = 0$: statically determinate structure ; $F < 0$: statically indeterminate (superstructure).
- Condition for a chain with only turning pairs having $F = 1$: $j = (3/2)\,n - 2$ and $l = 2p - 4$ (l links, p pairs). Minimum links in a chain = 4 ; the simplest is the four-bar.

Mechanism	n	j	h	F
Four-bar chain	4	4	0	$3(3)-8 = 1$
Single slider-crank	4	4	0	1
Five-bar chain	5	5	0	$3(4)-10 = 2$
Cam + roller follower	3	3	1	$3(2)-6-1 = -1 \rightarrow 1$ (idle roller)
Pantograph / Watt six-bar	6	7	0	$3(5)-14 = 1$

Ex 1. Check the mobility of a mechanism with 8 links and 10 turning pairs : $F = 3(8 - 1) - 2(10) = 21 - 20 = 1$ □ (single DOF). Ex 2. 6 links, 8 pairs : $F = 15 - 16 = -1 \Rightarrow$ structure (locked).

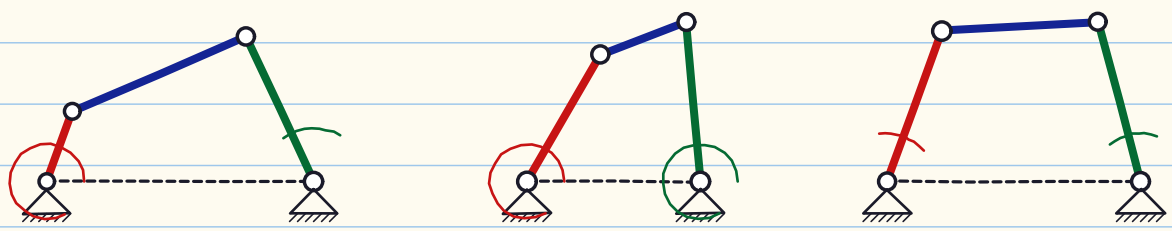
- **Redundant (idle) DOF** : motion that does not affect the output, e.g. rotation of a roller on a cam follower – subtract 1 from the formula. **Over-constrained** mechanisms (parallel links, special dimensions) move even though $F \le 0$ (e.g. parallelogram linkage, locomotive coupling rods).
- **Gruebler's for spatial** (Kutzbach 3-D) : $F = 6(n-1) - 5j_1 - 4j_2 - 3j_3 - 2j_4 - j_5$ (j_i = pairs with i constraints).

3. Four-Bar Chain & Grashof's Law

Crank – rocker
shortest link adjacent to fixed

Double crank
(drag link) – shortest fixed

Double rocker
coupler is shortest



$s + l \le p + q$ (Grashof) — $s + l > p + q$: all double rockers

A four-bar chain has four links (frame d, crank a, coupler b, follower c) connected by four turning pairs.

- Let s = shortest link, l = longest, p and q = other two. **Grashof's law** : for at least one link to make a full revolution relative to the others

$$s + l \leq p + q$$

- If $s + l < p + q$ (Grashof chain), the type depends on which link is fixed :

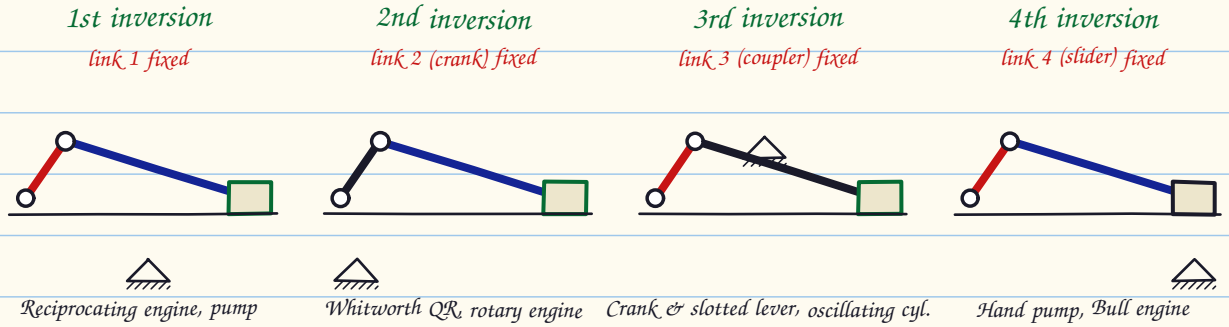
Fixed link	Mechanism	Example
Link adjacent to shortest	Crank-rocker (shortest = crank, other side rocker)	windscreen wiper, sewing machine treadle
Shortest link	Double crank (drag link)	locomotive coupling rods, parallel-ruler
Link opposite to shortest	Double rocker (shortest = coupler)	Watt straight line, Ackermann steering

- If $s + l > p + q$: non-Grashof – every inversion is a double rocker (only oscillation, no complete rotation). If $s + l = p + q$: "change point" – links become collinear, mechanism may flip (parallelogram linkage).

Ex 3. Links 30, 70, 80, 90 mm : $s + l = 30 + 90 = 120$; $p + q = 70 + 80 = 150$ $\Rightarrow 120 < 150 \Rightarrow$ Grashof ; fix 70 (adjacent to 30) → crank-rocker ; fix 30 → drag link ; fix 80 (opposite to 30) → double rocker.

- Special case : **Parallelogram linkage** ($a = c$, $b = d$) – opposite links equal : output rotates at the same speed as input (coupling rods of locomotive; pantograph). **Anti-parallelogram** (crossed) gives speed ratio ≠ 1.

4. Slider-Crank Chain & its Inversions



The single slider-crank chain : 3 turning pairs + 1 sliding pair. Four links : frame (link 1), crank (2), connecting rod (3), slider (4).

1st inversion – link 1 fixed

- Reciprocating engine and compressor : slider (piston) moves in cylinder, converts reciprocation ↔ rotation. Stroke = 2r (crank length r).

2nd inversion – link 2 (crank) fixed

- **Whitworth quick return** : crank fixed, slotted bar rotates fully, ram driven by link – used in shapers and slotters. **Rotary I.C. engine (Gnome)** : cylinders rotate about fixed crankshaft.

3rd inversion – link 3 (connecting rod) fixed

- **Oscillating-cylinder engine, crank & slotted lever quick return** (shaper), and the **slotted-lever** mechanism ; the slider oscillates / rotates about the fixed rod pivot.

4th inversion – link 4 (slider) fixed

- **Hand pump** (link 3 becomes the plunger rod), **Bull engine,** pendulum pump – rod oscillates in the fixed sleeve.

Double slider-crank chain – 2 sliding + 2 turning pairs

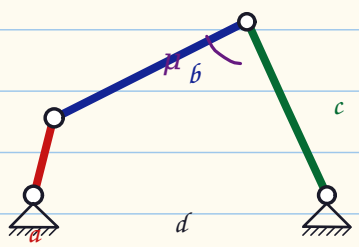
- Elliptic trammel : a point on the connecting bar traces an ellipse (draw ellipses). Scotch yoke : converts rotation to pure SHM of the yoke ($s = r \sin\theta$). Oldham's coupling : connects two parallel shafts with small offset – the intermediate disc slides in grooves, speed ratio 1 : 1.
- Other chains : **Pantograph** (copy to a scale, indicator mechanism), **Geneva (Maltese-cross)** – intermittent motion, **ratchet & pawl, toggle mechanisms.**

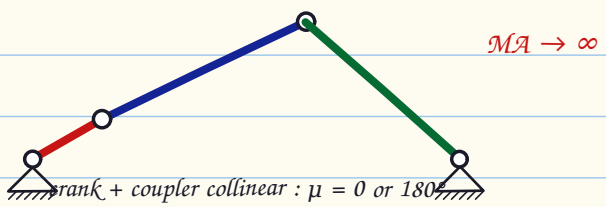
5. Limit Positions, Mechanical Advantage & Transmission Angle

Transmission angle μ

Limit (toggle) positions

$MA = c \sin\mu / (a \sin\delta)$





- **Limit (extreme / dead-centre) positions** : the crank and coupler become collinear (stretched or folded) – the driven link reaches its extreme position and the velocity of the output link is zero. For a slider-crank : inner and outer dead centres ; stroke $H = 2r$. For a crank-rocker the rocker's swing angle ψ is found from these two positions (cosine rule).
- **Transmission angle μ** : acute angle between the coupler and the output (follower) link ; measures the quality of force transmission. Best $\mu = 90°$; poor if $\mu < 40°$ (jamming, high bearing forces).

$$\cos\mu = [\, b^2 + c^2 - a^2 - d^2 + 2ad\cos\theta \,] / (2\,b\,c)\quad \mu_{min} \text{ at } \theta = 0,\ \mu_{max} \text{ at } \theta = 180°$$

- Recommended range $40° \le \mu \le 140°$. Pressure angle (for cams/sliders) $= 90° - \mu$.
- **Mechanical advantage** MA = output torque / input torque $= \omega_{in}/\omega_{out}$ (ideal, no friction). Since the coupler is a two-force member F acts along it : $T_4 = F\,c\sin\mu$, $T_2 = F\,a\sin\delta$ (δ = angle between crank and coupler) :

$$MA = T_4 / T_2 = (c \sin\mu) / (a \sin\delta)$$

- At toggle ($\delta = 0°$ or $180°$) $MA \to \infty$: very large force from small effort (stone crusher, toggle clamp, rock breaker) but motion of the output $\to 0$.

Ex 4. Four-bar $a = 50$, $b = 120$, $c = 80$, $d = 100$ mm. At $\theta = 0$: $\cos\mu = (120^2 + 80^2 - 50^2 - 100^2 + 2(50)(100))/(2\times120\times80) = (14400 + 6400 - 2500 - 10000 + 10000)/19200 = 18300/19200 = 0.953 \Rightarrow \mu = 17.6°$ – poor (μ_{min}) ; at $\theta = 180°$: $\cos\mu = (8300 - 10000)/19200 = -0.0885 \Rightarrow \mu = 95.1°$ (μ_{max}).

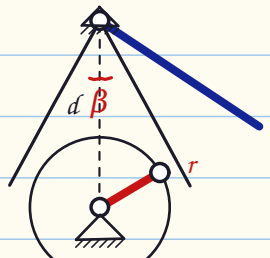
Stroke & time ratio of a slider-crank

- Time ratio (offset slider) $Q = (180° + \beta)/(180° - \beta)$, β = angle between the two lines of coupler at dead centres ($\beta = 0$ if no offset $\Rightarrow$ equal times). Stroke $H = 2r$ for in-line.

6. Quick-Return Mechanisms

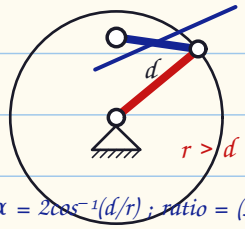
Used in shapers, slotters, presses – the cutting stroke should be slow and uniform, return stroke fast to save time.

Crank & slotted lever



$\sin(\beta/2) = r/d$; ratio = $(180°+\beta)/(180°-\beta)$

Whitworth quick return



$\alpha = 2\cos^{-1}(d/r)$; ratio = $(360°-\alpha)/\alpha$

cutting stroke : slow
return stroke : fast

Crank & slotted lever (3rd inversion)

- Fixed distance d between the crank pivot and the lever pivot, $d > r$. The lever swings between two tangent positions through angle β :

$$\sin(\beta/2) = r/d \quad \text{Time ratio} = \text{cutting time / return time} = (180° + \beta)/(180° - \beta)$$

- Length of stroke = 2 × (lever length to ram link) × $\sin(\beta/2)$ = L (ram arm) × (chord) ; ram stroke = $2R\sin(\beta/2)$ where R = length of the swinging arm (pivot to ram pin).

Ex 5. r = 100 mm, d = 250 mm : $\sin(\beta/2) = 0.4 \Rightarrow \beta = 47.2°$; time ratio = 227.2/132.8 = 1.71.

Whitworth quick return (2nd inversion)

- Crank fixed (O) is the fixed link of length d, driving crank length $r > d$. Lever pivot P at distance d from O ; slotted lever makes full revolution.

$$\alpha = 2\cos^{-1}(d/r) \quad \text{Time ratio} = (360° - \alpha)/\alpha = \text{cutting angle / return angle}$$

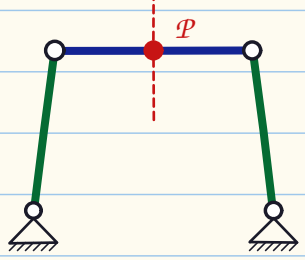
Ex 6. r = 150 mm, d = 100 mm : $\alpha/2 = \cos^{-1}(0.667) = 48.2° \Rightarrow \alpha = 96.4°$; ratio = (360 − 96.4)/96.4 = 2.73.

- Other QR : **Drag-link mechanism** (double-crank, shortest link fixed) – ratio up to 3 ; offset slider-crank.

7. Straight-Line Mechanisms

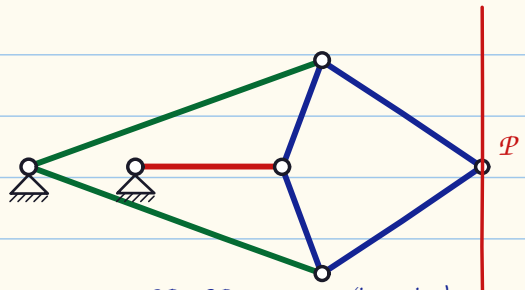
Convert rotary motion to straight-line motion using only lower pairs (no guides) – exact or approximate.

Watt (approximate)



P near mid-point of coupler

Peaucellier (exact straight line)

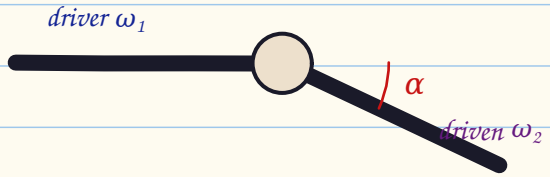


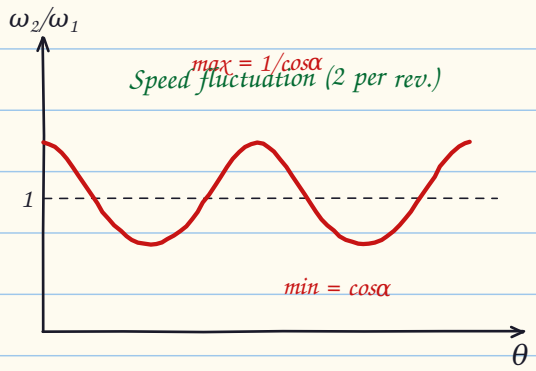
OB · OP = constant (inversion)

- **Exact** : Peaucellier (8 links – an inversor : OB · OP = const. so B moves on a circle through O, P on a straight line) ; Hart's (6 links, anti-parallelogram) ; Scott-Russell (a slider moves a bar with a mid-point fixed pivot – the end moves in a line perpendicular).
- **Approximate** : Watt (double rocker – P at the coupler mid-point moves on a figure-8 with a nearly straight part, used in steam engine indicator), Grasshopper, Tchebicheff, Robert's.
- Condition for the Peaucellier cell : $AB = BC = CD = DA$ (rhombus), $OA = OC$, and $OO_1 = O_1B$. Then $OB \times OD = OA^2 - AB^2$ = constant.
- Straight-line motion is needed in engine indicators, machine tools, sheet-metal handling and robotic arms.

8. Universal (Hooke's) Joint

Hooke's (universal) joint





A joint connecting two intersecting shafts (inclined at α) – used in automobile propeller shafts, rolling mills, machine tools. Two forks at right angles joined by a cross (spider) – a spherical four-bar linkage.

- If the driving shaft turns through θ and the driven through φ :

$$\tan\theta = \cos\alpha \cdot \tan\varphi \quad \omega_2/\omega_1 = \cos\alpha \,/\, (1 - \cos^2\theta \sin^2\alpha)$$

- Speed ratio varies twice per revolution : $\omega_{2,max} = \omega_1/\cos\alpha$ ($\theta = 0°, 180°$) ; $\omega_{2,min} = \omega_1\cos\alpha$ ($\theta = 90°, 270°$).

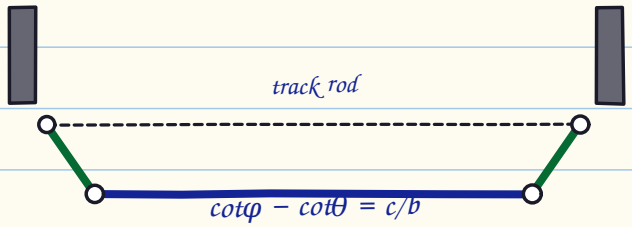
Coefficient of fluctuation of speed = $(\omega_{2max} - \omega_{2min})/\omega_1 = \tan\alpha \sin\alpha$; mean speed equal to ω_1.

- A larger α gives larger speed fluctuation, inertia torques and vibration ; practical $\alpha \le 15°–20°$.
- **Double Hooke's joint** : two joints with intermediate shaft ; if the driving & driven shafts make equal angles with the intermediate shaft and the forks of the intermediate shaft are in the same plane, $\omega_3 = \omega_1$ – constant-velocity transmission (automobile drive shaft, steering columns). Constant-velocity joints (Rzeppa, Tracta) also used.

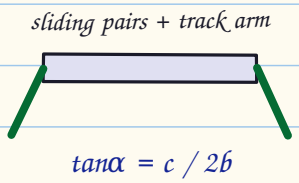
Ex 7. $\alpha = 20°$: $\omega_{2max} = \omega_1/0.9397 = 1.064\ \omega_1$; $\omega_{2min} = 0.9397\ \omega_1$; fluctuation = $\tan20° \sin20° = 12.4\ \%$. For $\omega_1 = 600$ rpm : 563.8 – 638.5 rpm.

9. Rocker Mechanisms & Steering Gears

Ackermann steering gear



Davis steering gear



- **Crank-rocker** – input crank rotates, output rocker oscillates : wipers, rocking-chair, radar scanners, looms. **Rocker-rocker** (double-rocker) – both oscillate : Ackermann steering, aircraft landing-gear. **Rocker-slider** – oscillating arm drives a slider : valve gear, shaper ram drive.
- **Steering gear condition** : for pure rolling of all wheels in a turn, axes of all wheels must meet at one point (instantaneous centre) – the inner wheel turns more than the outer :

$\cot\varphi - \cot\theta = c / b$ (φ outer, θ inner, c = distance between pivots, b = wheel base)

- **Davis steering gear** has sliding pairs and satisfies the condition exactly ($\tan\alpha = c/2b$) but wears quickly. **Ackermann** – only turning pairs, exact at straight-ahead and at one turning angle, approximately in between – the common choice.

Ex 8. Ackermann gear : wheel base $b = 2.4$ m, pivot distance $c = 1.2$ m. For inner wheel $\theta = 30°$: $\cot\varphi = \cot 30° + c/b = 1.732 + 0.5 = 2.232 \Rightarrow \varphi = 24.1°$ (outer wheel).

10. Other Common Mechanisms (short notes)

- **Pantograph** : four links forming a parallelogram with extended arms ; a point traces a path similar to another point to a fixed scale (ratio = OB/OA). Uses : copying drawings, engraving, indicator rigs, current collector on electric trains.
- **Geneva (Maltese-cross) drive** : a driving pin enters a slot of the driven wheel once per revolution and indexes it by $360°/n$; the wheel is locked in between by the circular disc – used in film projectors, indexing tables. For n slots the driver turns through $180° - 360°/n$ during indexing.
- **Ratchet & pawl** : intermittent rotation in one direction only (jacks, winches, ratchet spanner, clock escapement). **Toggle mechanism** : as the links become collinear a small effort gives very large force (stone crusher, riveting, clamps, punching).
- **Scotch yoke** displacement $x = r\sin\theta$, velocity $r\omega\cos\theta$, acceleration $-r\omega^2\sin\theta$ – pure SHM. **Rapson's slide** and **Corliss gear** are used in steam-ship steering and valve control.
- **Equivalent linkages** : a higher pair (cam-follower / gear contact) can be replaced for kinematic analysis by an equivalent lower-pair four-bar whose links join the centres of curvature of the mating profiles.
- **Kinematic diagram** : a skeleton of links drawn to scale showing only the dimensions that affect the motion – first step of every analysis (displacement, velocity, acceleration).

11. Additional Solved Problems

P1 (mobility). Plane mechanism with 7 links, 8 lower pairs and 1 higher pair : $F = 3(7-1) - 2(8) - 1 = 18 - 16 - 1 = 1$.

P2 (Grashof, change point). Links 40, 60, 100, 100 : $s + l = 140$, $p + q = 160 \Rightarrow s + l < p + q \Rightarrow$ Grashof □. Links 40, 60, 100, 120 mm : $s + l = 160 = p + q \Rightarrow$ change-point mechanism.

P3 (shaper design). Stroke of ram 250 mm, cutting : return ratio 1.8, crank & slotted lever. $(180 + \beta)/(180 - \beta) = 1.8 \Rightarrow 2.8\beta = 144 \Rightarrow \beta = 51.4°$; $\sin(\beta/2) = 0.434 \Rightarrow r/d = 0.434 \Rightarrow$ for $d = 300$ mm : $r =$ 130 mm ; swing arm $R = 250/(2 \times 0.434) =$ 288 mm.

P4 (slider-crank). Crank 120 mm, connecting rod 480 mm, in-line : stroke $= 2r = 240$ mm ; $n = l/r = 4$. Dead centres at distance $l + r = 600$ mm and $l - r = 360$ mm from the crank pivot (difference = stroke □).

P5 (transmission angle). A four-bar with $a = 20$, $b = 70$, $c = 60$, $d = 80$ mm : at $\theta = 0$, $\cos\mu = (4900 + 3600 - 400 - 6400 + 3200)/(2\times70\times60) = 4900/8400 = 0.583 \Rightarrow \mu_{min} =$ 54.3° ; at 180° : $\cos\mu = (1700 - 3200)/8400 = -0.179 \Rightarrow \mu_{max} = 100.3°$ (within 40°–140° □).

Short answers – Quick Revision

- **Difference between machine and mechanism?** A mechanism only transmits/modifies motion ; a machine also transforms energy and does work.
- **Why can a structure not be a mechanism?** $F \le 0$ – no relative motion possible between the links.
- **Why are higher pairs necessary?** Cams and gears give required non-uniform motion / speed ratios that lower pairs cannot, at the cost of high contact stress.
- **Why quick return needed?** Idle return stroke reduces non-productive time ; the cutting stroke takes the larger crank angle (speed uniform).
- **What is an inversion?** Fixing a different link of the same chain ⇒ different mechanism with the same relative motion between links.
- **Which inversion gives a slow cutting stroke and fast return?** Crank and slotted lever / Whitworth – 3rd and 2nd inversions of the single slider-crank.
- **Why is the double Hooke's joint used?** Cancels the speed fluctuation of a single joint ($\omega_1 = \omega_3$ when forks of intermediate shaft are in line).

12. Quick Formula Sheet – Unit I

Item	Formula
Mobility	$F = 3(n-1) - 2j_1 - j_2$
Grashof	$s + l \le p + q$
Transmission angle	$\cos\mu = (b^2+c^2-a^2-d^2+2ad\cos\theta)/2bc$

Item	Formula
Mechanical advantage	$MA = c \sin\mu / (a \sin\delta)$
Crank-slotted QR	$\sin(\beta/2) = r/d$; ratio $= (180+\beta)/(180-\beta)$
Whitworth QR	$\alpha = 2\cos^{-1}(d/r)$; ratio $= (360-\alpha)/\alpha$
Hooke's joint	$\tan\theta = \cos\alpha \tan\varphi$; $\omega_2/\omega_1 = \cos\alpha/(1-\cos^2\theta \sin^2\alpha)$
Steering	$\cot\varphi - \cot\theta = c/b$

13. Important Questions

- Define kinematic pair, link, kinematic chain, mechanism and inversion. Classify kinematic pairs with examples.
- Derive Grübler's criterion and find the DOF of the given linkages.
- State Grashof's law. Explain the inversions of the four-bar chain with sketches and applications.
- Explain the inversions of a single slider-crank chain and the double slider-crank chain.
- What is transmission angle? Derive the expression for mechanical advantage of a four-bar.
- Explain the working of Whitworth quick return and crank & slotted lever mechanisms; derive the time ratio.
- Explain the Peaucellier and Watt mechanisms. Derive the speed ratio of a Hooke's joint; double Hooke's joint.
- Explain Ackermann & Davis steering gears with the condition for correct steering.

UNIT – II

Plane Motion Analysis

Subject : Theory of Machines (23A03403)

Syllabus: Displacement, velocity & acceleration analysis of simple mechanisms, graphical velocity analysis using instantaneous centres, analysis using loop-closure equations, kinematic analysis of simple mechanisms, slider-crank dynamics, coincident points, Coriolis component of acceleration.

1. Introduction – Motion of a Link

Plane motion of a rigid link = translation + rotation. Two points A and B of the same rigid link keep a constant distance, so the velocity of B relative to A is perpendicular to AB :

$$v_{BA} = \omega \times AB \ (\perp AB) \quad v_B = v_A + v_{BA}$$

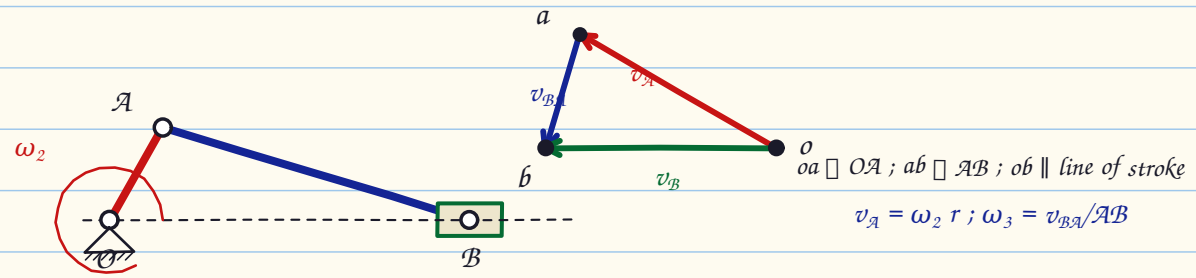
- **Acceleration** of B relative to A has two components : radial (centripetal) $a^r_{BA} = \omega^2 \, AB = v_{BA}^2/AB$ (directed from B toward A) and tangential $a^t_{BA} = \alpha \cdot AB$ ($\perp AB$).

$$a_B = a_A + a^r_{BA} + a^t_{BA}$$

- Angular velocity of a link = v_{BA}/AB (relative velocity of any two points ÷ distance) ; angular acceleration $\alpha = a^t_{BA}/AB$.
- A point on a rotating crank of radius r has $v = \omega r$, $a_n = \omega^2 r$ (toward the centre), $a_t = \alpha r$.
- **Methods of analysis** : (i) graphical – relative velocity (velocity polygon), instantaneous centre, Klein's construction ; (ii) analytical – loop-closure vector equations, complex numbers, matrices ; (iii) computer / numerical.

2. Displacement, Velocity & Acceleration of a Slider-Crank

Configuration diagram Velocity polygon (relative velocity)



Crank OA = r, connecting rod AB = l, n = l/r = obliquity ratio, crank angle θ measured from inner dead centre (IDC), φ = obliquity angle of connecting rod : sinφ = sinθ/n.

$$x = r\,(1 - \cos\theta) + l\,(1 - \cos\varphi) \approx r\,[\,(1 - \cos\theta) + \sin^2\theta/(2n)\,]$$ (piston displacement from IDC)

- Piston velocity : $v = r\,\omega\,(\sin\theta + \sin 2\theta/(2n)) = r\omega\,\sin(\theta+\varphi)/\cos\varphi$ (exact).
- Piston acceleration : $a = r\,\omega^2\,(\cos\theta + \cos 2\theta/n)$ ($a = d\,v/dt$, $\alpha_{crank} = 0$).
- Angular velocity of connecting rod $\omega_3 = \omega\,\cos\theta\,/\sqrt{(n^2 - \sin^2\theta)}$; ω_3 is max at θ = 0 and 180°, zero at the dead centres of the rod (θ = 90°, 270° for large n).
- Acceleration of the piston is maximum at the dead centres : $a = r\omega^2(1 + 1/n)$ at IDC (θ = 0) and $-r\omega^2(1 - 1/n)$ at ODC ; velocity is maximum at about θ = 75°–80° (n = 4 : 1.03 rω, slightly more than the crank-pin speed). For $n \to \infty$ the motion becomes SHM ($v = r\omega\,\sin\theta$, $a = r\omega^2\cos\theta$).

θ (deg)	x/r	v/(rω)	a/(rω²)	φ (deg)
0	0	0	+1.250	0
30	0.165	0.609	+0.991	7.2
60	0.595	0.977	+0.375	12.5
90	1.127	1.000	−0.250	14.5
120	1.595	0.755	−0.625	12.5
150	1.897	0.391	−0.741	7.2
180	2.000	0	−0.750	0

Table for n = l/r = 4 (θ from IDC). Note the piston covers more than half its stroke (x/r = 1.127 at 90°) in the first 90° of crank – asymmetry due to the finite rod

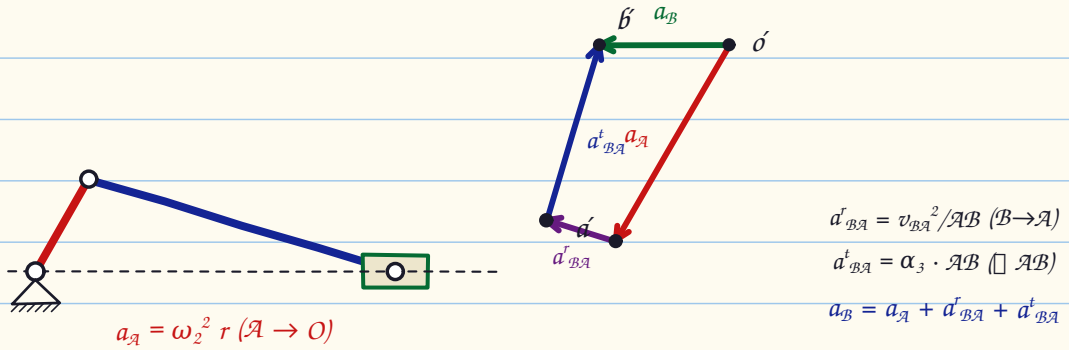
length.

> Ex 1. $r = 100$ mm, $l = 400$ mm ($n = 4$), $N = 300$ rpm ($\omega = 31.42$ rad/s), $\theta = 45°$: $v = 0.1 \times 31.42\,(0.7071 + 1/8) = 2.61$ m/s (exact 2.62) ; $a = 0.1(31.42)^2(0.7071 + 0) = 69.8$ m/s² ; $\omega_3 = 31.42 \times 0.7071/\sqrt{(16 - 0.5)} = 5.64$ rad/s.

Relative velocity (velocity polygon) method – steps

- (1) Draw the configuration diagram to scale. (2) $v_A = \omega_2\, r \perp OA$ – draw oa. (3) From a draw ab $\perp$ AB (direction of v_{BA}) ; from o draw ob along the line of stroke ; intersection = b. (4) v_B = ob, v_{BA} = ab, ω_3 = ab/AB. (5) The velocity of any point C on AB : locate c on ab such that ac/ab = AC/AB (velocity image).
- Angular velocity sense : v_{BA} direction relative to A shows rotation of AB (CW/CCW).

$\alpha_2 = 0$ (uniform crank speed) Acceleration polygon



Acceleration polygon (graphical)

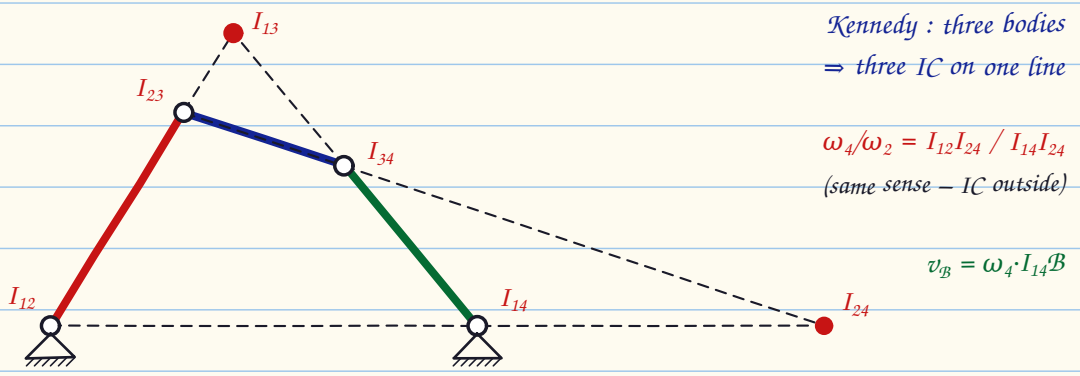
- $a_A = \omega_2^2\, r$, directed A → O (no tangential part for constant ω_2). Draw o′a′. From a′ draw $a^r_{BA} = (ab)^2/AB$ parallel to BA, direction B → A. From its end draw a perpendicular line (tangential component). From o′ draw the line parallel to the slider path – intersection b′ ⇒ a_B = o′b′ . $\alpha_3 = a^t_{BA}/AB$.
- **Klein's construction** : a graphical shortcut on the configuration diagram itself. Extend the connecting rod to cut the perpendicular through O to the line of stroke at M ; then OM represents the piston velocity ($v_B = \omega \cdot OM$) and AM the velocity of B relative to A. A second set of auxiliary circles on OA and AB gives the acceleration vectors without drawing a separate polygon.

3. Instantaneous Centre Method (graphical velocity)

Instantaneous centre (IC) of two bodies in relative plane motion = the point (of both bodies) that has zero relative velocity – at that instant body 2 appears to rotate about it.

- Number of IC : $\mathcal{N} = n(n-1)/2$ (n = number of links). Four-bar → 6 ; slider-crank → 6. Types : **fixed** (I_{12}, I_{14} at frame pins), **permanent** (pin joints between moving links I_{23}, I_{34}), **neither fixed nor permanent** (I_{13}, I_{24}).
- **Kennedy's (Aronhold–Kennedy) theorem** : the three instantaneous centres of three bodies in relative motion lie on a straight line.

Instantaneous centres of a four-bar ($\mathcal{N} = 6$)



Locating the centres (four-bar)

- I_{12}, I_{23}, I_{34}, I_{14} = pin centres. I_{13} = intersection of lines $I_{12}I_{23}$ and $I_{14}I_{34}$. I_{24} = intersection of $I_{12}I_{14}$ (frame line) and $I_{23}I_{34}$ (coupler line).
- Slider-crank : I_{12} = O, I_{23} = A, I_{34} = B, I_{14} = at infinity ⊥ to the slider path ; I_{13} = intersection of OA (extended) and the perpendicular to the slider path through B ; I_{24} at infinity (moves ∥ to the path).
- Rolling contact : the IC is at the point of contact. Sliding contact : the IC lies on the common normal at the point of contact.

Velocity from IC

$$v = \omega \times (\text{distance from IC}) \quad \omega_4/\omega_2 = (I_{12}I_{24}) / (I_{14}I_{24})$$

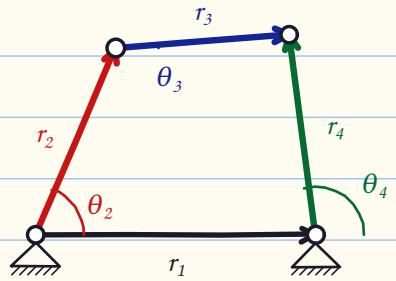
- I_{24} is common to links 2 and 4 so its velocity can be computed from either : $v = \omega_2 \cdot I_{12}I_{24} = \omega_4 \cdot I_{14}I_{24}$. If I_{24} lies between I_{12} and I_{14} the rotation senses are opposite, if outside the same.

Ex 2. (slider-crank above, $\theta = 45°$) : $x_B = r\cos\theta + \sqrt{(l^2 - r^2\sin^2\theta)} = 70.7 + 393.7 = 464.4$ mm ; $I_{13}B = x_B \tan\theta = 464.4$ mm ; $OI_{13} = x_B/\cos\theta = 656.8$ mm $\Rightarrow I_{13}A = 556.7$ mm. $\omega_3 = v_A/I_{13}A = 3.1416/0.5567 =$ 5.64 rad/s ; $v_B = \omega_3 \cdot I_{13}B = 5.64 \times 0.4644 =$ 2.62 m/s.

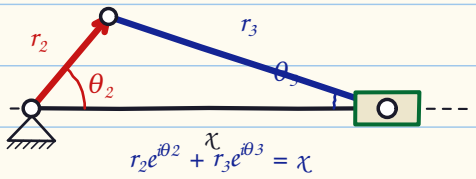
- Advantages : quick for velocity of a point ; disadvantages : not for acceleration ; poor accuracy when centres are far away.

4. Loop-Closure (Vector-Loop) Equations

Four-bar vector loop



Slider-crank vector loop



Treat each link as a vector $r_i e^{i\theta_i}$; the sum of vectors round a closed loop is zero. Differentiate for velocity and acceleration (analytic, exact, computer-ready).

(a) Slider-crank (in-line)

$$r_2\cos\theta_2 + r_3\cos\theta_3 = x \qquad r_2\sin\theta_2 + r_3\sin\theta_3 = 0$$

- θ_3 (angle of the rod vector A→B) is negative : $\sin\theta_3 = -(r_2/r_3)\sin\theta_2$.
- **Velocity** (differentiate w.r.t. time) : $\omega_3 = -r_2\omega_2\cos\theta_2 / (r_3\cos\theta_3)$; $\dot{x} = -r_2\omega_2\sin\theta_2 - r_3\omega_3\sin\theta_3$.
- **Acceleration** : $\alpha_3 = [\, r_2\omega_2^2\sin\theta_2 + r_3\omega_3^2\sin\theta_3 \,] / (r_3\cos\theta_3)$; $\ddot{x} = -r_2\omega_2^2\cos\theta_2 - r_3\alpha_3\sin\theta_3 - r_3\omega_3^2\cos\theta_3$ ($\alpha_2 = 0$).

Ex 3. $r_2 = 0.1$, $r_3 = 0.4$, $\theta_2 = 45°$, $\omega_2 = 31.42$ rad/s : $\sin\theta_3 = -0.1768$ ($\theta_3 = -10.18°$) ; $\omega_3 = -0.1(31.42)(0.7071)/(0.4 \times 0.9843) =$ −5.64 rad/s (clockwise, opposite to crank) ; $\dot{x} = -2.221 - 0.4(-5.64)(-0.1768) =$ −2.62 m/s ; $\alpha_3 = (69.8 - 2.25)/0.3937 = 171.5$ rad/s² ; $\ddot{x} = -69.8 + 12.1 - 12.5 =$ −70.2 m/s² (toward O).

Ex 3(a). Four-bar : $r_1 = 100$, $r_2 = 40$, $r_3 = 90$, $r_4 = 60$ mm ; $\theta_2 = 60°$, $\omega_2 = 10$ rad/s ($\alpha_2 = 0$). Solving the loop : $\theta_3 = 16.1°$, $\theta_4 = 83.8°$. Velocity matrix gives $\omega_3 = -1.94$ rad/s, $\omega_4 = 4.99$ rad/s ⇒ $v_A = 0.40$ m/s, $v_B = r_4\omega_4 = 0.30$ m/s. Acceleration : $\alpha_3 = 27.5$ rad/s², $\alpha_4 = 47.8$ rad/s².

(b) Four-bar linkage

$$r_2 e^{i\theta_2} + r_3 e^{i\theta_3} - r_4 e^{i\theta_4} - r_1 = 0$$

- *Real and imaginary parts give two equations.* ***Position*** *(Freudenstein) with input θ_2, output θ_4 :*

$$K_1 \cos\theta_4 - K_2 \cos\theta_2 + K_3 = \cos(\theta_2 - \theta_4)\quad K_1 = r_1/r_2,\ K_2 = r_1/r_4,\ K_3 = (r_2^2 - r_3^2 + r_4^2 + r_1^2)/(2r_2r_4)$$

- ***Velocity*** *(differentiate) : $r_2\omega_2\, e^{i\theta_2} + r_3\omega_3\, e^{i\theta_3} - r_4\omega_4\, e^{i\theta_4} = 0$ ⇒ in matrix form*

$$[\, -r_3\sin\theta_3 \;\; r_4\sin\theta_4 \;;\; r_3\cos\theta_3 \;\; -r_4\cos\theta_4 \,]\ \{\, \omega_3,\ \omega_4 \,\} = r_2\omega_2\ \{\, \sin\theta_2,\ -\cos\theta_2 \,\}$$

- ***Acceleration*** *: differentiate again – the tangential terms (α_3, α_4) are the unknowns and ω^2 terms are on the right-hand side. Solve with Cramer's rule or numerically (Newton-Raphson).*
- *Advantages : accuracy, complete cycle in one program ; extends to any linkage, synthesis and optimisation.*

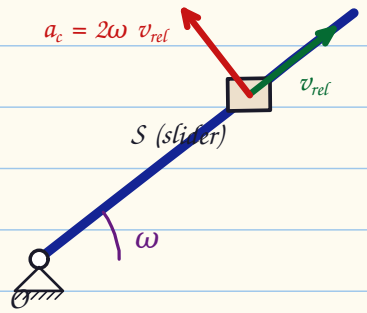
Procedure for a general mechanism

- *(1) Choose independent loops (number = $j - n + 1$). (2) Write vector-loop equations with angles measured from a fixed axis. (3) Solve position for unknown angles (closed-form or iterative). (4) Differentiate once – linear in velocities. (5) Differentiate twice – linear in accelerations.*

5. Coincident Points & Coriolis Acceleration

Coincident points *: two points of different links that occupy the same position at an instant (e.g. a slider on a rotating link and the point of the link under it).*

Coriolis component



Slider on a rotating link

$a_S = a_P + a_{S/P}^{rel} + 2\ \omega \times v_{rel}$

P = coincident point on link

Direction : rotate v_{rel} through 90° in the direction of ω

Exists when : $v_{rel} \neq 0$ and $\omega \neq 0$

Let P be the point on the rotating link 2 (angular velocity ω) and S the slider on it (relative velocity $v_{S/P} = v_{rel}$ along the link). Then

$$v_S = v_P + v_{rel}\ a_S = a_P + a_{rel} + a_c$$

$$a_c\ (Coriolis) = 2\ \omega\ v_{rel}$$

- Components of a_S : (1) a_P (radial $\omega^2 r$ and tangential αr of the link point), (2) a_{rel} = relative acceleration along the slot (sliding), (3) Coriolis $a_c = 2\omega v_{rel}$, perpendicular to the slot, direction obtained by rotating v_{rel} through 90° in the direction of ω.
- Origin : the sliding velocity changes direction due to rotation of the link, while the link's point velocity changes magnitude because the slider moves to a larger radius – two equal contributions (ωv_{rel} each).
- It exists only when the slider/point has a relative velocity on a rotating link ($v_{rel} \neq 0$, $\omega \neq 0$). Zero when $\omega = 0$ (translation) or $v_{rel} = 0$ (rigid) – so none in the simple slider-crank (slider moves in a fixed guide).
- Examples : crank & slotted lever, Whitworth quick return, oscillating-cylinder engine, Geneva, cam with sliding roller, aircraft in a turn, Foucault's pendulum, cyclones, ocean currents.

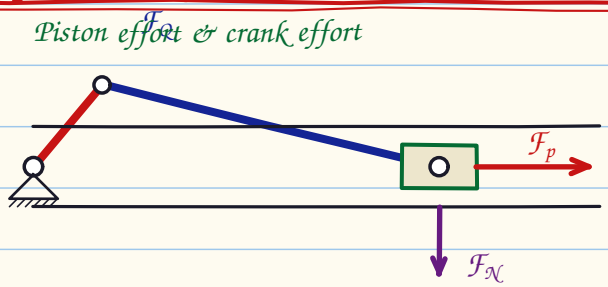
Ex 4. A slider moves outward in a radial slot of a link rotating at $\omega = 5$ rad/s (constant) ; at radius 0.3 m, $v_{rel} = 0.4$ m/s, $a_{rel} = 0$. Absolute speed : $v = \sqrt{((0.3\times5)^2 + 0.4^2)} = 1.55$ m/s. Acceleration : radial = $\omega^2 r = 7.5$ m/s² (toward centre) ; Coriolis = $2(5)(0.4) = 4$ m/s² ⊥ slot ⇒ $a = \sqrt{(7.5^2 + 4^2)} = 8.5$ m/s².

Procedure for a quick-return (crank & slotted lever)

- Crank pin A (on link 2) and the slotted-lever point A' (coincident, on link 3 = lever) : $v_A = v_{A'} + v_{A/A'}$ (sliding along lever) ⇒ gives lever angular velocity $\omega_3 = v_{A'}/(PA)$.
- Acceleration : a_A (radial, crank) $= a_{A'}^{\,r} + a_{A'}^{\,t} + a_c + a_{rel}$. Solve the polygon for α_3 and a_{rel}.

6. Dynamics of the Slider-Crank Mechanism

Piston effort & crank effort



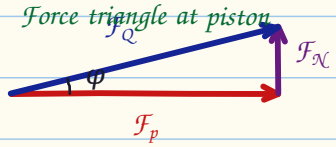
$\sin\varphi = \sin\theta / n$, $n = l/r$

$F_N = F_p \tan\varphi$

$F_Q = F_p / \cos\varphi$

$T = F_p\, r \sin(\theta+\varphi)/\cos\varphi$

Force triangle at piston



Forces on the piston and crank

- Piston (gas) force $F = (\pi/4)D^2\, p$ (net of friction). Inertia of reciprocating parts (mass m) :

$$F_I = m\, r\, \omega^2\, (\cos\theta + \cos 2\theta/n) \text{ primary} = mr\omega^2\cos\theta \text{ , secondary} = mr\omega^2\cos 2\theta/n$$

- **Net (piston) effort** $F_p = F - F_I - mg$ (vertical engines add weight ; sign for inertia opposes acceleration).
- Thrust on the cylinder wall $F_N = F_p \tan\varphi$; thrust along the connecting rod $F_Q = F_p/\cos\varphi$; force on the crank pin = F_Q.
- **Crank effort (turning moment)** :

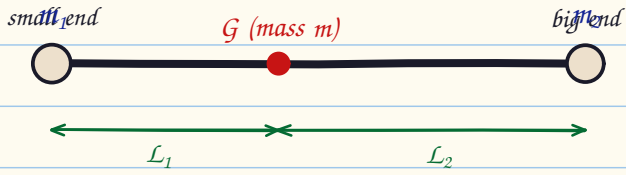
$$T = F_Q\, r \sin(\theta+\varphi) = F_p\, r \sin(\theta+\varphi)/\cos\varphi \approx F_p\, r\, (\sin\theta + \sin 2\theta/(2\sqrt{(n^2 - \sin^2\theta)}))$$

- Radial component $F_R = F_Q \cos(\theta+\varphi)$ loads the crankshaft bearings ; reaction torque on the frame = T (equal & opposite) ⇒ tends to overturn the engine : T = F_N × (crank-shaft height).
- Mean torque from the indicator diagram ; inertia causes torque fluctuations (flywheel – Unit V) and shaking force F_I on the frame (balancing – Unit IV).

Ex 5. $m = 20$ kg reciprocating mass, $r = 0.1$ m, $N = 300$ rpm, $n = 4$, $\theta = 45°$: $F_I = 20(0.1)(31.42)^2(0.7071 + 0) =$ **1396 N** (acts opposite to piston acceleration). If gas force = 12 kN : $F_p = 12000 - 1396 = 10604$ N ; $\sin\varphi = 0.1768 \Rightarrow \varphi = 10.2°$; $F_N = 10604 \tan 10.2° =$ **1908 N** ; $T = 10604 \times 0.1 \times \sin(55.2°)/\cos 10.2° =$ **883 N·m**.

Dynamically equivalent system of connecting rod

Dynamically equivalent system of the connecting rod



1) $m_1 + m_2 = m$

2) $m_1L_1 = m_2L_2$

3) $m_1L_1^2 + m_2L_2^2 = I_G$

($K^2 = L_1L_2$)

The distributed mass of the connecting rod is replaced by two point masses (at small end & big end) so that inertia effects are equal :

- Three conditions : (1) total mass equal $m_1 + m_2 = m$; (2) CG unchanged $m_1L_1 = m_2L_2$; (3) moment of inertia equal $m_1L_1^2 + m_2L_2^2 = mk_G^2$.
- Condition 3 requires $L_1L_2 = k_G^2$. If only two masses are placed at the ends ($L_1 + L_2 = l$) condition 3 is not satisfied – a correction **torque** $T_c = m(L_1L_2 - K^2)\alpha$ must be added. Usual : 1/3 mass at crank pin, 2/3 at piston pin (approx.).
- Inertia of the equivalent system : m_2 (small end) adds to the reciprocating mass ; m_1 (big end) is a rotating mass at the crank-pin radius.

8. Special Topics in Velocity Analysis

Velocity image & velocity of a point on a link

- For a point C on link AB, in the velocity polygon the line ab is the "image" of AB : point c divides ab in the same ratio as C divides AB (ac/ab = AC/AB). For a link with three points the triangle abc is similar to ABC and rotated 90° (**velocity image principle**).
- Velocity of a point on a link given its two other points : draw the image first, join the pole o to c ⇒ v_C. Needed for centre of gravity velocity in dynamics.

Velocity of rubbing at a pin joint

- Rubbing velocity at a pin of radius r between two links : $(\omega_1 \pm \omega_2)\, r$ — '+' when the links rotate in opposite directions, '−' when in the same direction. Used to find the work lost in friction at joints and wear.
- At a revolute pair of a slider-crank : crank pin $(\omega_2 - \omega_3) r_p$; crosshead pin $\omega_3\, r_c$; crank shaft $\omega_2\, r_s$.

Klein's construction (slider-crank)

- Extend the connecting rod AB to meet the perpendicular through O to the line of stroke at M. Triangle OAM is similar to the velocity triangle $\Rightarrow v_B \propto OM$, $v_{BA} \propto AM$, and (for constant ω) a_B is obtained from the intersections of auxiliary circles : draw a circle on AB as diameter and a circle with centre A, radius AB^2/OA..., the quadrilateral gives $a_B = \omega^2 \times$ (length). Saves the polygon.

Mechanisms with sliders on rotating links — using IC

- Crank & slotted lever : I_{13} (crank-lever) lies on the line of the crank pin and the lever pivot ; the velocity of the sliding block relative to the slot is found by resolving v_A along and perpendicular to the lever.
- Scotch yoke : $x = r\sin\theta \Rightarrow v = r\omega\cos\theta$, $a = -r\omega^2\sin\theta$; the IC of the yoke is at infinity (pure translation).

9. Additional Solved Problems

P1 (IC ratio). Four-bar with $I_{12}I_{24} = 0.20$ m and $I_{14}I_{24} = 0.08$ m (I_{24} outside the frame pins $\Rightarrow$ same sense), $\omega_2 = 20$ rad/s : $v_{I24} = 20 \times 0.20 = 4$ m/s $\Rightarrow \omega_4 = 4/0.08 =$ **50 rad/s** ($\omega_4/\omega_2 = 2.5$).

P2 (relative velocity). A rod $AB = 0.5$ m has $v_A = 3$ m/s along the rod's perpendicular-free direction ; if $v_B = 5$ m/s and $v_A \perp v_B$: $v_{BA} = \sqrt{(5^2 - 3^2)} = 4$ m/s $\perp AB \Rightarrow \omega_{AB} = 4/0.5 =$ **8 rad/s**.

P3 (piston speed). Engine $r = 75$ mm, $l = 300$ mm, 1200 rpm : $\omega = 125.7$ rad/s ; at $\theta = 90°$ $v = r\omega(1 + 0) = 9.42$ m/s ($r\omega\sin\theta + r\omega\sin 180°/2n = 9.42$) ; mean piston speed $= 2\,L\,N/60 = 2(0.15)(1200)/60 =$ **6.0 m/s**.

P4 (max acceleration). Same engine : a_{max} (IDC) $= r\omega^2(1 + 1/n) = 0.075(125.7)^2(1 + 0.25) =$ **1481 m/s²** ; at ODC $r\omega^2(1 - 1/n) = 889$ m/s².

P5 (Coriolis in QR). Lever rotating at $\omega = 4$ rad/s, slider moving along it at $v_{rel} = 0.6$ m/s : $a_c = 2 \times 4 \times 0.6 = 4.8$ m/s².

Common mistakes

- Direction of a^r_{BA} is $B \to A$ (towards the reference point) ; tangential is $\perp AB$, sense consistent with α.
- Forgetting the Coriolis term for a slider on a rotating link or using ω of the wrong link (use ω of the link carrying the slot).
- Mixing rpm and rad/s ; using degrees in calculator mode for sin/cos of radians.

10. Quick Formula Sheet – Unit II

Item	Formula
Relative motion	$v_{BA} = \omega \cdot AB$; $a^r_{BA} = \omega^2 AB$; $a^t_{BA} = \alpha \cdot AB$
Piston	$v = r\omega(\sin\theta + \sin 2\theta/2n)$; $a = r\omega^2(\cos\theta + \cos 2\theta/n)$
Rod angular velocity	$\omega_3 = \omega \cos\theta / \sqrt{(n^2 - \sin^2\theta)}$
Instantaneous centres	$N = n(n-1)/2$; $\omega_4/\omega_2 = I_{12}I_{24}/I_{14}I_{24}$
Loop closure	$\Sigma\, r_i e^{i\theta_i} = 0$; differentiate for v, a
Coriolis	$a_c = 2\omega\, v_{rel}$
Inertia force	$F_I = mr\omega^2(\cos\theta + \cos 2\theta/n)$
Forces	$F_N = F_p \tan\varphi$; $F_Q = F_p/\cos\varphi$; $T = F_p\, r \sin(\theta+\varphi)/\cos\varphi$

11. Important Questions

- Explain the relative-velocity method; draw velocity and acceleration polygons for a slider-crank mechanism and derive piston velocity & acceleration.
- State Kennedy's theorem; locate all instantaneous centres of a four-bar and slider-crank mechanism ; find ω_4.
- Write the loop-closure equations for a four-bar and a slider-crank and obtain the velocity and acceleration equations.
- What are coincident points? Derive the Coriolis component of acceleration and state its direction rule.
- Obtain the piston effort, thrust on the cylinder wall, force along the connecting rod and crank effort of a slider-crank.

- Explain the dynamically equivalent two-mass system of a connecting rod.

UNIT – III

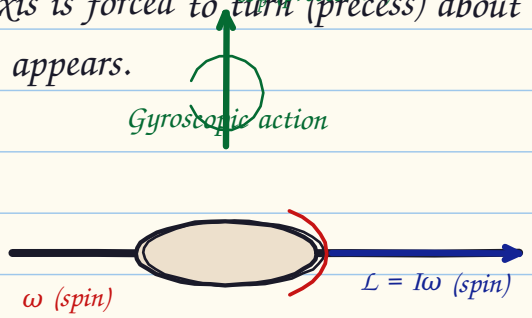
Gyroscope & Gear Profile

Subject : Theory of Machines (23A03403)

Syllabus: Principle of gyroscope, gyroscopic effect in aeroplane, ship, car & two-wheeler ; involute & cycloidal profiles, gear parameters, fundamental law of gearing & conjugate action, contact ratio, interference/undercutting, helical, bevel, worm, rack & pinion, epicyclic & regular gear trains.

1. Gyroscope – Principle

A gyroscope is a spinning rotor mounted so that its axis can turn (gimbals). When the spin axis is forced to turn (precess) about an axis perpendicular to it, a gyroscopic couple appears.

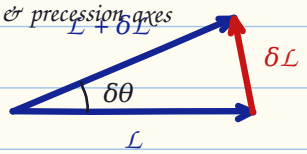


$\delta L = L\,\delta\theta = I\omega\,\delta\theta$

$dL/dt = I\omega\,\omega_p$

Active couple $C = I\omega\,\omega_p$

⊥ to both spin & precession axes



Reaction couple on frame = $-C$

- Spin velocity ω (rad/s) about the spin axis ; precession (angular velocity of the axis) ω_p about the perpendicular axis ; moment of inertia of rotor I.
- Angular momentum $L = I\omega$ (vector along spin axis by right-hand rule). In time δt the spin axis turns by $\delta\theta \Rightarrow$ change in momentum $\delta L = L\,\delta\theta$ (perpendicular to L).

$$C = dL/dt = I\,\omega\,\omega_p \text{ (active gyroscopic couple)}$$

- Direction : the active couple vector is $\omega_p \times L$ (right-hand rule) – the axis tends to turn so that spin vector moves towards the precession vector. The **reaction couple** on the supporting frame (bearings) is equal and opposite.

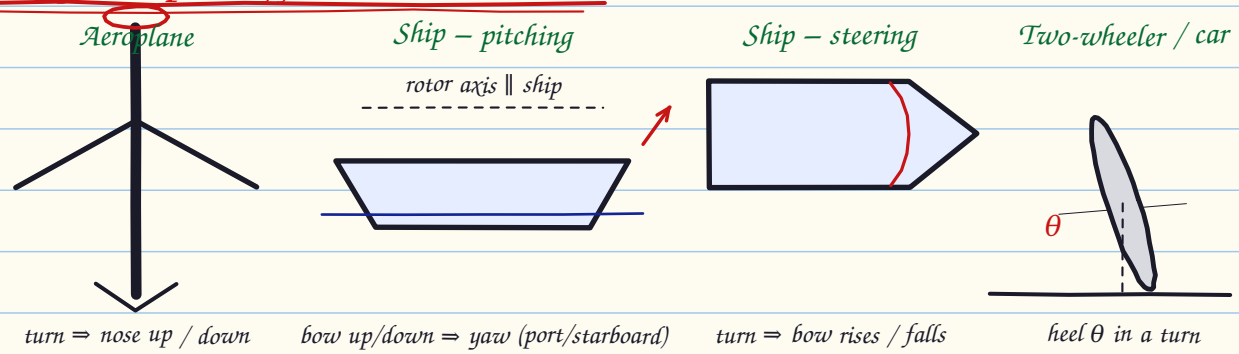
- Units : I in kg·m² = $m\,k^2$; ω_p in rad/s ; C in N·m. For harmonic precession (pitching of ship) $\omega_{p,max} = (2\pi/T)\,\varphi_{max}$ (φ_{max} = amplitude in rad, T = period).

Rules to find the reaction couple

- Draw the spin vector (right-hand screw). Draw the precession vector. Active couple = turn the spin vector towards the precession vector through 90° ; reaction couple = reverse.
- When the spin axis and precession axis are parallel (rolling of a ship), no gyroscopic couple. If they are inclined at angle θ : $C = I\omega\omega_p \sin\theta$.
- Applications : gyrocompass, gyro-stabiliser for ships, aircraft attitude indicators, inertial navigation, spinning top, bicycle stability, projectile (rifling), gyro-seismometer.

Ex 1. Ship rotor mass 3000 kg, k = 0.6 m, 1500 rpm ; ship pitches simple harmonically with amplitude 6° and period 20 s : I = 3000 × 0.36 = 1080 kg·m² ; ω = 157.1 rad/s ; $\omega_{p,max} = (2\pi/20)(6\pi/180) = 0.0329$ rad/s ⇒ C_{max} = 1080 × 157.1 × 0.0329 = **5580 N·m**.

2. Gyroscopic Effects on Vehicles



(a) Aeroplane

- Rotating parts (engine, propeller) have spin about the longitudinal axis. When the aeroplane **turns** (yaw precession, $\omega_p = v/R$ about the vertical) the reaction couple acts in the pitching plane (nose up/down) :

$$C = I\,\omega\,\omega_p = I\,\omega\,(v/R)$$

- Rule (propeller rotating clockwise viewed from the rear) : turn left ⇒ nose rises (tail falls) ; turn right ⇒ nose falls. For anticlockwise rotation the effects are reversed.

- Pitching of the aircraft (nose up/down) produces a couple about the vertical axis (tends to yaw it) ; rolling produces none (axes parallel).

> Ex 2. Engine + propeller $I = 36\ kg{\cdot}m^2$, 2400 rpm clockwise from rear ; aeroplane flying at 400 km/h turns right on a 600 m radius. $\omega = 251.3$ rad/s ; $v = 111.1$ m/s $\Rightarrow \omega_p = 0.1852$ rad/s ; $C = 36 \times 251.3 \times 0.1852 =$ 1676 N·m ; it tends to depress the nose and raise the tail (right turn, clockwise).

(b) Ship

- Rotor (turbine) axis along the ship's length. Three motions : **steering** (yaw), **pitching** (up-down oscillation of bow), **rolling** (no effect).
- Steering : same as aeroplane – with rotor clockwise looking from the stern, turning left raises the bow ; turning right lowers the bow (effect is a couple in the vertical plane).
- Pitching (bow moving up) ⇒ reaction couple yaws the ship : for a clockwise rotor viewed from the stern, upward pitching ⇒ ship tends to turn to starboard (right) and downward pitching ⇒ to port (left).
- Effect on bearings : reaction couple resisted by a pair of reactions $\pm C/L$ at the bearings (L = distance between bearings). For pitching of period T, maximum angular velocity occurs at the mean position.

(c) Two-wheeled vehicle taking a turn

- Wheels spin about horizontal axis ; during a turn at radius R the vehicle precesses at $\omega_p = v/R$. If the vehicle is inclined at θ to the vertical, the effective precession component is $\omega_p \cos\theta$ ⇒

$$C_g = (I_{w1}\omega_{w1} + I_{w2}\omega_{w2} \pm I_e\omega_e) (v/R) \cos\theta$$

- Engine rotor sign : '+' if it rotates in the same direction as wheels, '–' if opposite. The reaction gyroscopic couple tends to overturn the vehicle outwards (away from the centre).
- Centrifugal force $F_c = m v^2/R$ acts at height h of the CG : centrifugal couple = $(mv^2/R)\, h \cos\theta$ (also outward). The weight couple $mgh \sin\theta$ must balance them :

$$\tan\theta = v^2/(gR) + C_{g0}/(m\, g\, h)\ (C_{g0} = \text{gyro couple for } \theta = 0)$$

Ex 3. Motor cycle + rider 200 kg, $h = 0.6$ m, $v = 36$ km/h (10 m/s), $R = 30$ m. Wheel $I = 0.4$ kg·m² each, wheel radius 0.3 m ($\omega_w = 33.3$ rad/s) ; engine rotor $I_e = 0.2$ kg·m² at 5 × wheel speed, same direction. $C_{g0} = (2\times0.4\times33.3 + 0.2\times166.7)(10/30) = 20$ N·m ; $\tan\theta = 100/(9.81\times30) + 20/(200\times9.81\times0.6) = 0.340 + 0.017 = 0.357 \Rightarrow \theta = 19.6°$ to the vertical.

(d) Four-wheeled vehicle (car) taking a turn

- Three effects on the vertical reactions of the wheels : (i) weight mg/2 on each side ; (ii) **gyroscopic couple** from wheels & engine $C_g = (4I_w\omega_w \pm I_e\omega_e G)(v/R)$, producing $\Delta P = C_g/(2a)$ on outer wheels (+) and inner (–) where 2a = track width ; (iii) **centrifugal couple** $(mv^2/R)h \Rightarrow \Delta P = mv^2h/(2aR)$.

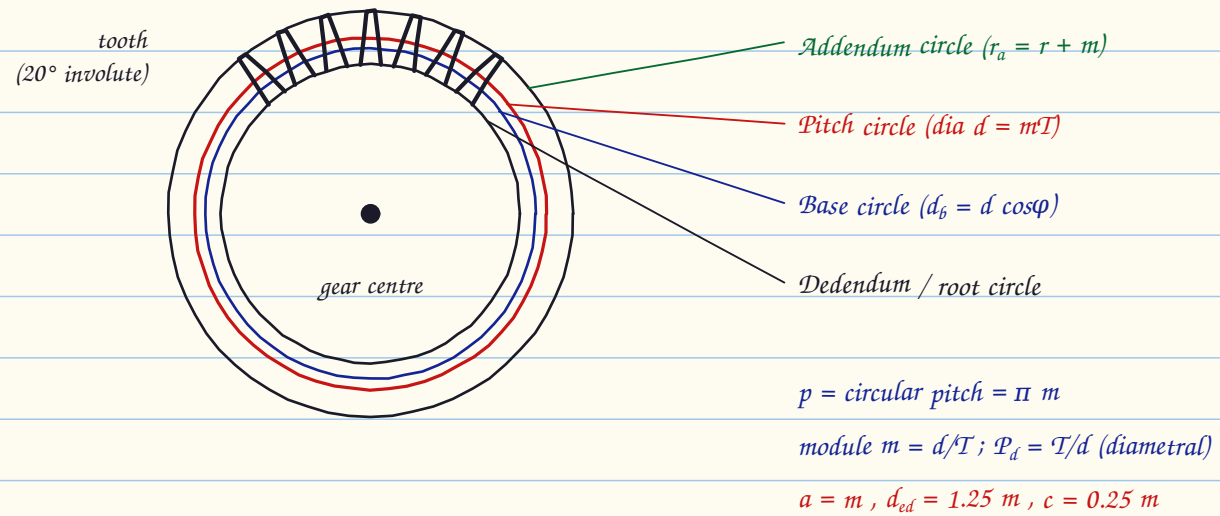
$$P_{outer} = mg/2 + (1/2a)[\, C_g + m\,v^2\,h\,/R\,] \qquad P_{inner} = mg/2 - (1/2a)[\, C_g + m\,v^2\,h/R\,]$$

- Limiting speed for overturning (inner wheels lift off, $P_{inner} = 0$) ; skidding when $F_c > \mu\, mg$ $(v^2/R > \mu g)$ – whichever occurs first.
- Stability remedy : lower CG, wider track, bank the road, opposite-rotating engine rotor (gyro compensation), tyres with high μ.

3. Gear Profile – Basics

Gears transmit motion and power between shafts with *positive drive, constant velocity ratio and compact size*. Gear types : spur (parallel shafts), helical, herringbone, bevel (intersecting), worm / hypoid / crossed-helical (skew), rack & pinion.

Spur gear terminology



Spur gear terminology

- **Pitch circle** : imaginary circle rolling without slip ; diameter d. **Module** $m = d/T$ (mm) – standard 1, 1.25, 1.5, 2, 2.5, 3, 4, 5, 6, 8, 10… ; **diametral pitch** $P_d = T/d$; **circular pitch** $p = \pi d/T = \pi m$; $p\,P_d = \pi$.
- **Addendum** $a = m$ (distance pitch → tip) ; **dedendum** $1.25\,m$; **clearance** $c = 0.25\,m$; **whole depth** $2.25\,m$; **working depth** $2m$. **Tooth thickness** = space width = $p/2$ (no backlash) ; backlash = space − thickness.
- **Pressure angle** φ : angle between common normal and common tangent at the pitch point ($14½°$, $20°$, $25°$). **Base circle** $d_b = d\cos\varphi$. **Base pitch** $p_b = p\cos\varphi$. **Face width** b (≈ 8–12 m). **Centre distance** $C = (d_1 + d_2)/2 = m(T_1 + T_2)/2$.
- **Velocity ratio** $VR = \omega_1/\omega_2 = N_1/N_2 = d_2/d_1 = T_2/T_1$ (external gears – opposite rotation).

Fundamental law of gearing (conjugate action)

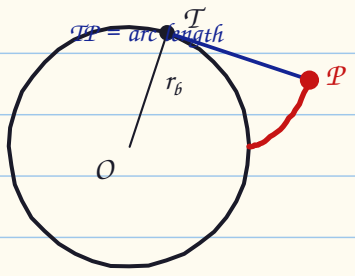
For a constant velocity ratio the common normal at the point of contact of the mating teeth must always pass through a fixed point P (the pitch point) on the line of centres :

$$\omega_1/\omega_2 = O_2P / O_1P = r_2/r_1$$

- Profiles that satisfy this law are called **conjugate profiles.** Commonly used : involute and cycloidal.

Involute profile

Generation of an involute



Properties

- TP (tangent) is the normal to the curve at P
- involute starts at the BASE circle ; none inside
- shape depends only on r_b ($= r\cos\varphi$)
- inv $\varphi = \tan\varphi - \varphi$; $r = r_b/\cos\varphi$
- centre distance change ⇒ ratio unchanged

- Involute = curve traced by the end of a string unwound from the base circle. Tooth profile from the base circle to the addendum circle. The common normal = tangent to base circles = the **line of action** – a straight line inclined at φ to the common tangent through P.

- Properties : (i) normal to the involute is tangent to the base circle ; (ii) the involute ratio depends only on base radii ⇒ a small change of centre distance does not affect the velocity ratio ; (iii) pressure angle constant during mesh (single line of action) ⇒ constant force direction and smooth running ; (iv) easy manufacture (straight-sided rack cutter, hobbing). Involute function $\text{inv}\,\varphi = \tan\varphi - \varphi$.

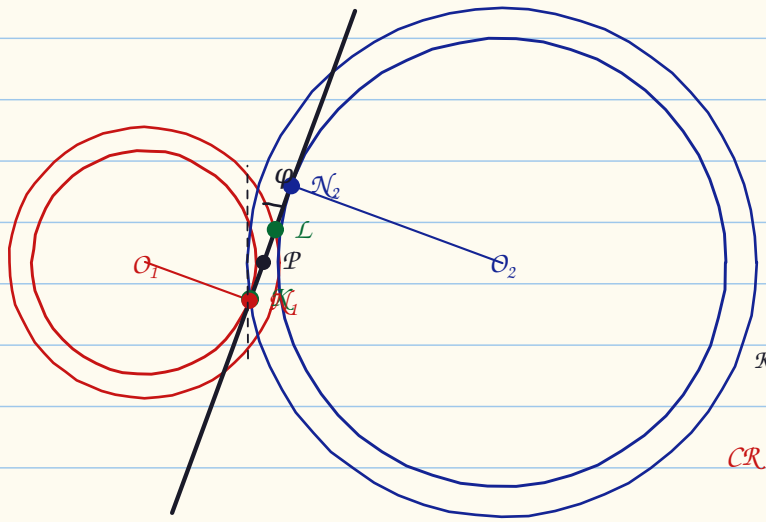
Cycloidal profile

- Face = **epicycloid** (rolling circle outside pitch circle) and flank = **hypocycloid** (rolling inside).

Item	Involute	Cycloidal
Profile	single curve face & flank	epicycloid + hypocycloid
Pressure angle	constant	varies (zero at pitch point)
Centre-distance change	ratio unchanged	ratio changes – exact distance needed
Interference	possible (small T)	none
Wear, strength	flank weaker at root	stronger, less wear (convex-concave)
Manufacture	easy, cheap	difficult
Use	almost all gears	clock & watch gears, heavy-load low-speed

4. Path of Contact, Arc of Contact & Contact Ratio

Path of contact $K L$, line of action $N_1 N_2$



KP : path of approach

PL : path of recess

$CR = KL / (\pi\, m \cos\varphi)$

- **Path of contact** KL : locus of the contact point – part of the line of action between the addendum circles. K = where the addendum circle of the **driven** gear cuts the line of action (start), L = where the addendum circle of the driver cuts it (end).
- With R_{a1}, R_{a2} = addendum radii and R_{b1}, R_{b2} = base radii, φ = pressure angle, C = centre distance :

$$KP = \sqrt{(R_{a2}^2 - R_{b2}^2)} - R_2 \sin\varphi \qquad PL = \sqrt{(R_{a1}^2 - R_{b1}^2)} - R_1 \sin\varphi$$

$$\text{Path of contact } KL = \sqrt{(R_{a1}^2 - R_{b1}^2)} + \sqrt{(R_{a2}^2 - R_{b2}^2)} - C \sin\varphi$$

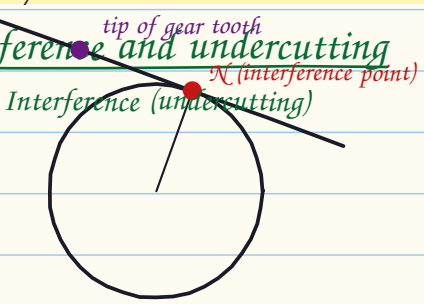
- **Arc of contact** = $KL / \cos\varphi$. **Contact ratio** (average number of teeth in contact) :

$$CR = \text{arc of contact} / \text{circular pitch} = KL / (\pi\, m \cos\varphi) = KL / p_b$$

- CR must be > 1 ($\geq$ 1.2 practically ; 1.4–1.8 for spur) so that a new pair engages before the previous leaves – otherwise noise & impact. A CR of 1.6 means 2 pairs in contact 60 % of time and 1 pair 40 %.

Ex 4. m = 5 mm, T_1 = 20, T_2 = 40, φ = 20°, addendum = m. R_1 = 50, R_2 = 100, R_{a1} = 55, R_{a2} = 105, R_{b1} = 46.98, R_{b2} = 93.97, C = 150 mm. $KP = \sqrt{(105^2 - 93.97^2)} - 100 \sin20° = 46.85 - 34.20 = 12.65$; $PL = \sqrt{(55^2 - 46.98^2)} - 50 \sin20° = 28.59 - 17.10 = 11.49$; KL = 24.14 mm ⇒ arc = 25.69 mm ; CR = 25.69/15.71 = 1.64.

Interference and undercutting



Occurs when the tip of the driven gear tooth cuts the pinion flank below the base circle

Avoid by : $T \geq T_{min} = 2A_w/\sin^2\varphi$ (rack)

- raise φ (20°, 25°)
- stub teeth
- profile shift
- more teeth on pinion
- reduce gear tip radius

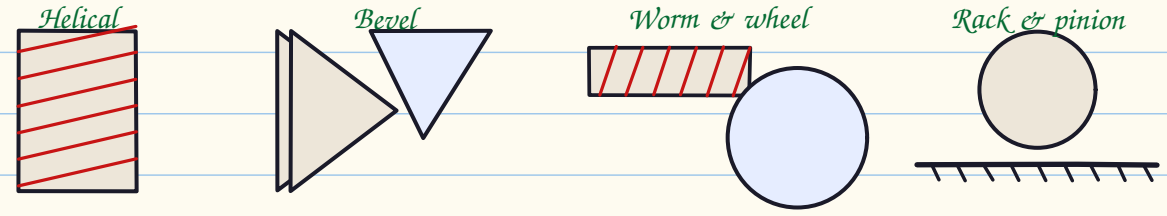
- **Interference** : tip of a tooth of one gear contacts the non-involute portion (below base circle) of the mating tooth, so the teeth jam/cut. It occurs when the addendum circle of the gear cuts the line of action beyond the tangent point N_1. **Undercutting** : the cutter removes the root of the pinion tooth ⇒ weak tooth.

- **Minimum number of teeth** on the pinion (gear ratio $G = T_G/T_P$, addendum A_w m) :

$$T_{min} = 2A_w / [\sqrt{(G^2 + (1 + 2G)\sin^2\varphi)} - G] \text{ (rack, } G \to \infty : T_{min} = 2A_w/\sin^2\varphi)$$

- For $\varphi = 20°$, $A_w = 1$: rack → 17.1 (18 practically 17), $G = 1$ → 12.3 ⇒ 13, $G = 2$ → 14.2 ⇒ 15. For 14½° → 32 teeth ; for 25° → 12 teeth.
- **Remedies** : increase pressure angle, stub teeth ($A_w < 1$), more teeth on pinion, profile shifting (addendum modification / X-gears), reduce gear addendum, adjust centre distance.

5. Types of Gears



Helical gears

- Teeth cut at a helix angle ψ ; contact starts at one end and spreads gradually – smoother, quieter, higher load capacity and contact ratio ; but produce axial thrust $F_a = F_t \tan\psi$ (use herringbone / double helical to cancel).
- Normal module $m_n = m\cos\psi$ (m = transverse module) ; pitch dia $d = T\, m_n/\cos\psi$; **virtual number of teeth** $T_v = T/\cos^3\psi$ (used for strength and interference). Shafts parallel (opposite hand) or crossed (same hand). Minimum teeth lower than spur.

> Ex 5. Helical : $T = 30$, $m_n = 4$ mm, $\psi = 25° \Rightarrow m = 4/0.9063 = 4.41$ mm ; $d = 132.4$ mm ; $T_v = 30/0.9063^3 = 40.3$.

Bevel gears

- Intersecting shafts (usually 90°). Pitch cones roll without slip ; pitch cone angles δ_1, δ_2 with $\tan\delta_1 = T_1/T_2$ ($\Sigma = 90°$) ; types : straight, spiral, zerol, hypoid. Virtual (Tredgold) teeth $T_v = T/\cos\delta$.

> Ex 6. $T_1 = 20$, $T_2 = 40$, shafts 90° : $\delta_1 = \tan^{-1}(0.5) = 26.6°$, $\delta_2 = 63.4°$; $T_{v1} = 20/0.894 = 22.4$; $T_{v2} = 40/0.447 = 89.4$.

Worm and worm wheel

- Worm (screw) with Z starts drives the wheel with T_G teeth ; shafts skew (usually 90°) ; velocity ratio = T_G/Z (single reduction up to 100 : 1). Lead $L = Z \times$ axial pitch ; lead angle $\tan\lambda = L/(\pi\, d_w)$.
- Efficiency $\eta = (\cos\varphi_n - \mu\, \tan\lambda)/(\cos\varphi_n + \mu\, \cot\lambda)$; **self-locking** when $\lambda <$ friction angle ($\eta < 50\ \%$) – used in hoists, lifts, steering gear. Sliding action ⇒ heat, bronze wheel, oil.

Rack & pinion

- Rack = gear of infinite radius (straight pitch line, straight sides) : $v = \omega\, r$, linear travel = $\pi\, d$ (per rev) ; used in lathe carriage, steering, rack railway, jacks, drilling machines.

6. Gear Trains

Simple, compound and reverted trains

- **Simple** : one gear per shaft ; idler changes direction only. Speed ratio = T_{last}/T_{first} (idlers cancel). **Compound** : two or more gears on one shaft :

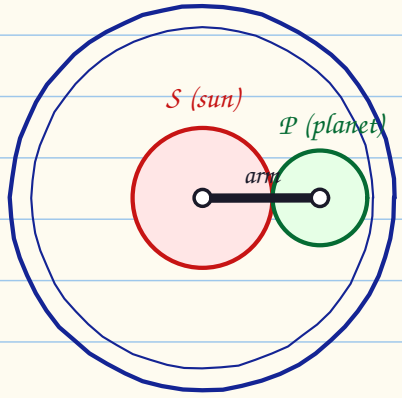
$$N_{last}/N_{first} = (T_1T_3\ldots)/(T_2T_4\ldots)\ \text{(product of drivers / product of followers)}$$

- **Reverted** : first and last shafts coaxial – centre distances equal : $m(T_1 + T_2) = m(T_3 + T_4)$ (lathe back gear, clocks).

Ex 7. $A(20)$ drives $B(50)$; $C(15)$ on B's shaft drives $D(60)$: speed ratio $N_A/N_D = (50/20)(60/15) = 10$; if $N_A = 1000$ rpm ⇒ $N_D = 100$ rpm (same direction – two meshes).

Epicyclic (planetary) gear trains

Epicyclic (planetary) gear train



R (ring / annulus) $T_R = T_S + 2T_P$

Tabular method (arm A, sun S, planet P, ring R)

Condition	Arm	Sun	Planet	Ring
1 Arm fixed, S turns +1	0	+1	$-T_S/T_P$	$-T_S/T_R$
2 Add +y to all	y	y	y	y
Total	y	y+1	$y - T_S/T_P$	$y - T_S/T_R$

Ex: ring fixed ⇒ $y = T_S/T_R$ ... see notes

- Axes of one or more gears move about the axis of the sun gear – carried by the **arm.** Compact, high ratio in small space, two DOF (one input fixed ⇒ 1 DOF). Uses : automatic transmission, differential, hoists, speed reducers, clocks.
- **Tabular method** : (1) arm fixed, give sun +1 turn ; find relative turns of other gears (external mesh reverses sign, internal mesh keeps sign) ; (2) add +y turns to all (arm turn y) ; (3) apply given conditions (a gear fixed or known speed) and solve for y.

Ex 8. Sun S(20), planet P(20), ring R(60 = 20 + 2×20). Ring fixed, sun input 1000 rpm : row 1 : arm 0, S +1, P −1, R −1/3 ; row 2 (add y) : arm y, S y + 1, P y − 1, R y − 1/3. R fixed ⇒ y = 1/3 ⇒ S = 4/3. Ratio S/arm = 4 ⇒ arm speed = 1000/4 = 250 rpm (same direction) ; planet = (1/3 − 1) × 750 = −500 rpm.

Ex 9. Simple epicyclic : gear A (40 teeth) fixed, gear B (30 teeth) meshing and carried by the arm, arm 100 rpm : B = arm $(1 + T_A/T_B)$ = 100(1 + 40/30) = 233 rpm in the same direction.

- **Differential** of an automobile : bevel epicyclic ; on a straight road both axles turn at the cage speed ; in a turn $N_L + N_R = 2\,N_{cage}$.
- Torque : the sum of the external torques on the train is zero ⇒ $T_{in} + T_{out} + T_{fixed} = 0$; power loss small ⇒ $T_{in}\,\omega_{in} \approx T_{out}\,\omega_{out}$.

7. Additional Solved Problems

P1 (gyro car). A car wheel (I_w = 1.0 kg·m², rolling radius 0.35 m) at 72 km/h (20 m/s) rounds a 50 m curve : ω_w = 57.1 rad/s ; ω_p = 0.4 rad/s ; C_{wheel} = 1.0 × 57.1 × 0.4 = 22.9 N·m per wheel ; four wheels 91.4 N·m (equal in direction, effect shared on outer/inner wheels).

P2 (gear pair). m = 4 mm, T_1 = 24, T_2 = 60 ⇒ d_1 = 96, d_2 = 240, C = 168 mm ; $p = \pi m$ = 12.57 mm ; VR = 2.5 ; $d_{b1} = 96 \cos 20° = 90.2$ mm.

P3 (minimum teeth). Pinion & gear equal (G = 1), φ = 20°, full depth : $T_{min} = 2/(\sqrt{(1 + 3 \sin^2 20°)} - 1) = 2/0.1623 = 12.3 \Rightarrow$ 13 teeth.

P4 (worm). Single-start worm, T_G = 40 ⇒ ratio 40 : 1. If worm turns 1200 rpm ⇒ wheel 30 rpm. Lead angle for d_w = 50 mm, axial pitch 12.57 mm : $\tan\lambda = 12.57/(\pi \times 50) = 0.080 \Rightarrow \lambda = 4.6°$ (self-locking if $\mu > 0.08$).

P5 (contact ratio). Path of contact KL = 20 mm, m = 4 mm, φ = 20° : $p_b = \pi m \cos\varphi$ = 11.81 mm ⇒ CR = 20/11.81 = 1.69 (about 2 pairs in contact for 69 % of the time).

Short answers – Quick Revision

- **Why does a ship's rolling cause no gyroscopic effect?** The roll axis is parallel to the rotor spin axis – no precession perpendicular to the spin vector.
- **Why is involute preferred?** Constant pressure angle and velocity ratio insensitive to centre-distance errors ; easy to cut.
- **Why do helical gears run quieter?** Gradual engagement along the face ; higher contact ratio.
- **How to increase the contact ratio?** More teeth, lower pressure angle, longer addenda.
- **When is an epicyclic train preferred?** High reduction/compact coaxial layout, and variable speed ratio by holding different members.

8. Quick Formula Sheet – Unit III

Item	Formula
Gyroscopic couple	$C = I\omega\omega_p$; $\omega_p = v/R$
Aeroplane / ship	$C = I\,\omega\, v/R$; ship pitching $\omega_{p,max} = (2\pi/T)\varphi_{max}$
Two-wheeler heel	$\tan\theta = v^2/gR + C_{g0}/(mgh)$
Gear	$m = d/T$; $p = \pi m$; $d_b = d\cos\varphi$; $C = m(T_1+T_2)/2$

Item	Formula
Fundamental law	$\omega_1/\omega_2 = O_2P/O_1P$
Path of contact	$KL = \sqrt{(R_{a1}^2 - R_{b1}^2)} + \sqrt{(R_{a2}^2 - R_{b2}^2)} - C\sin\varphi$
Contact ratio	$CR = KL/(\pi\, m\cos\varphi)$
Min. teeth	$T_{min} = 2A_w/(\sqrt{(G^2+(1+2G)\sin^2\varphi)} - G)$
Helical	$T_v = T/\cos^3\psi$; $m_n = m\cos\psi$
Bevel	$\tan\delta_1 = T_1/T_2$; $T_v = T/\cos\delta$
Worm	$VR = T_G/Z$; $\tan\lambda = L/\pi d_w$

9. Important Questions

- Explain gyroscopic couple; derive $C = I\omega\omega_p$. Explain the effect on aeroplane, ship (steering, pitching, rolling), two-wheeler and car taking a turn.
- State and prove the fundamental law of gearing. Compare involute and cycloidal profiles.
- Define path of contact, arc of contact and contact ratio ; derive expressions. What is interference? Derive the minimum number of teeth on a pinion.
- Describe helical, bevel, worm gears and rack & pinion with applications. Define virtual number of teeth.
- Explain simple, compound, reverted and epicyclic gear trains ; solve epicyclic problems by the tabular method.

UNIT – IV

Balancing of Rotating Masses & Cams

Subject : Theory of Machines (23A03403)

Syllabus: Need for balancing, balancing of a single mass and several masses in different planes (analytical & graphical) ; classification of cams & followers, terminology, displacement diagrams – uniform velocity, parabolic, SHM, cycloidal ; derivatives of follower motion ; specified-contour cams – circular and tangent cams ; pressure angle & undercutting.

1. Balancing – Need

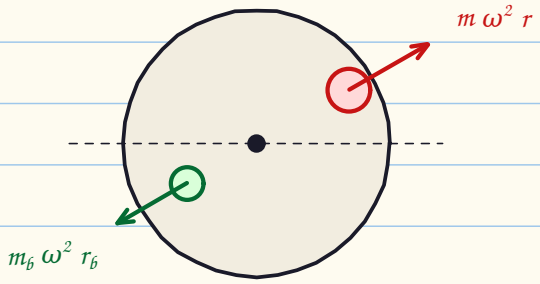
A rotating mass m at radius r with angular velocity ω produces a centrifugal force $F = m\,\omega^2\,r$ which acts radially and rotates with the shaft – it causes vibration, noise, bearing wear, fatigue failure and discomfort.

- Example : 5 kg unbalanced mass at $r = 0.1$ m, $N = 3000$ rpm : $\omega = 314.2$ rad/s $\Rightarrow F = 5 \times 0.1 \times 314.2^2 = 49.3$ kN ($\approx$ 1000 times its weight) – hence balancing of rotors of turbines, motors, crankshafts, wheels, fans.
- **Static balance** : resultant centrifugal force = 0 (the CG lies on the axis) $\Rightarrow \Sigma\, m\,r = 0$ (vector sum). **Dynamic balance** : resultant force = 0 and resultant couple = 0 $\Rightarrow \Sigma\, m\,r = 0$ and $\Sigma\, m\,r\,l = 0$. A dynamically balanced rotor is always statically balanced, not vice versa (e.g. two equal masses at opposite sides in different planes).
- Causes of unbalance : eccentric mounting, non-uniform material, keyways, bent shafts, unequal blades, wear. Balancing = adding/removing mass so that the inertia forces are equal and opposite.

2. Single Rotating Mass

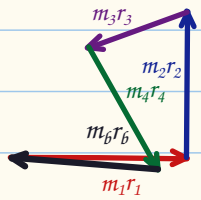
(a) Balanced by one mass in the same plane

Single mass in one plane



$m_b r_b = m r$ (opposite)

Force polygon (same plane)



closing vector = balancing mr

Mass m at radius r. Add a balancing mass m_b at radius r_b, diametrically opposite, in the same plane :

$$m_b r_b = m r$$

(b) Balanced by two masses in two other planes

A single mass m in plane A at radius r cannot be balanced by one mass in a different plane (forms a couple). Use two masses m_1, m_2 in planes 1 and 2 at distances l_1, l_2 from A :

- Force condition : $m_1r_1 + m_2r_2 = m r$ (both on the opposite side).
- Couple condition (moments about plane 2) : $m r \times l_2 = m_1r_1 (l_1 + l_2) \Rightarrow m_1r_1 = m r\ l_2/(l_1 + l_2)$; likewise $m_2r_2 = m r\ l_1/(l_1 + l_2)$ — like the reactions of a simply-supported beam. If A lies between the planes both balancing masses are opposite to m ; if A is outside, the nearer mass is on the same side as m.
- **Principle** : a mass in one plane can be replaced by an equal mass in the reference plane plus a couple (m r × distance).

3. Several Masses in the Same Plane

- Masses m_1, m_2, ... at radii r_1, r_2, ... and angles θ_1, θ_2, ... (all in one plane) ⇒ unbalanced force vector Σ m r. **Graphical** : draw the polygon of vectors m r (head-to-tail, parallel to the mass directions) ; the closing vector, reversed, is the balancing $m_b r_b$ (see figure).
- **Analytical** : resolve into components :

$\Sigma H = \Sigma\, m\, r \cos\theta$ $\Sigma V = \Sigma\, m\, r \sin\theta$ $R = \sqrt{(\Sigma H^2 + \Sigma V^2)}$, $\tan\theta_b = \Sigma V / \Sigma H$ (balance at $\theta_R + 180°$)

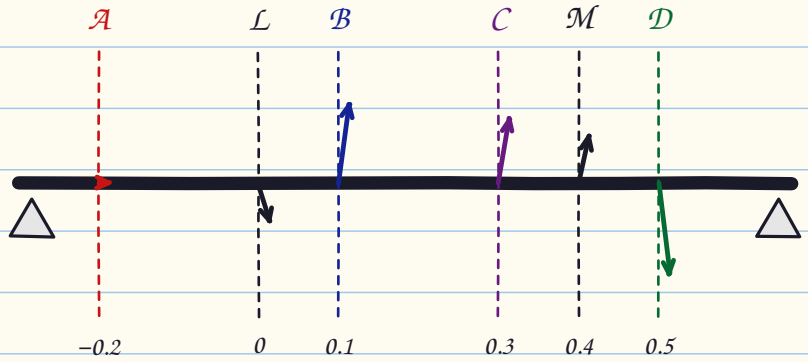
- Balancing mass $m_b = R / r_b$, placed at $\theta_b = \theta_R + 180°$.

Ex 1. Masses 15 kg at 0.2 m (0°), 10 kg at 0.25 m (90°), 12 kg at 0.15 m (200°), 8 kg at 0.3 m (300°). mr : 3.0, 2.5, 1.8, 2.4 kg·m. $\Sigma H = 3.0 + 0 - 1.69 + 1.2 = 2.51$; $\Sigma V = 0 + 2.5 - 0.62 - 2.08 = -0.19 \Rightarrow R = 2.52$ kg·m at −4.4°.

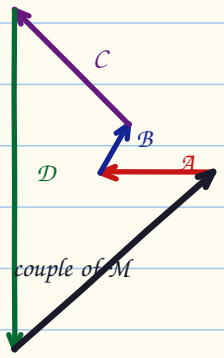
Balancing $mr = 2.52$ kg·m at 175.6° ; if $r_b = 0.2$ m $\Rightarrow m_b = 12.6$ kg.

4. Several Masses in Different Planes

Several planes : reference plane L
Planes (distances from L in m) — L, M = balancing planes



Couple polygon ($m\,r\,\chi$)



closing vector $\Rightarrow m_M r_M \chi_M$

Masses in different transverse planes along the shaft : both conditions (force & couple) must be satisfied. A reference plane (RP) is chosen ; the balancing masses are placed in two planes L and M (one of which may be chosen as RP to simplify).

Analytical procedure

- (1) Choose the reference plane (usually L) ; tabulate for each mass : m, r, θ, mr, distance l from RP (+ve to the right, −ve to left), couple mrl.
- (2) **Couple balance** : resolve mrl components : $\Sigma H_c = \Sigma\, mrl \cos\theta$, $\Sigma V_c = \Sigma\, mrl \sin\theta \Rightarrow$ the balancing couple of the mass in plane M : $m_M r_M l_M = \sqrt{(\Sigma H_c^2 + \Sigma V_c^2)}$ acting opposite to the resultant couple.
- (3) **Force balance** : include $m_M r_M$; $\Sigma H + m_M r_M \cos\theta_M + m_L r_L \cos\theta_L = 0$ and similarly for $V \Rightarrow m_L r_L$ and θ_L.
- Taking moments about the plane L eliminates the unknown mass in L.

Graphical procedure

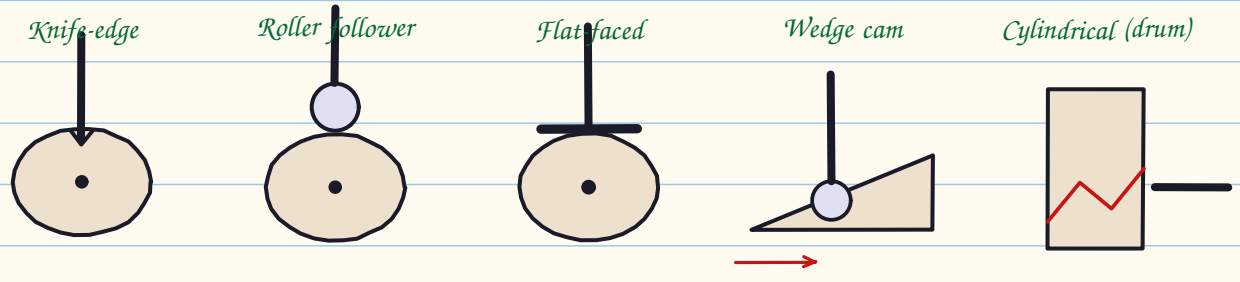
- Draw the couple polygon (vectors m r l, drawn parallel to the radii) ⇒ the closing vector gives $m_M r_M l_M$; draw the force polygon including $m_M r_M$ ⇒ closing vector gives $m_L r_L$ and its angle. (Couple vectors are drawn in the direction of the mass if l is positive, reversed if negative.)

Ex 2. A (200 kg, 0.15 m, 0°, l = −0.2 m), B (300 kg, 0.10 m, 60°, +0.1), C (240 kg, 0.12 m, 135°, +0.3), D (260 kg, 0.14 m, 270°, +0.5) about plane L ; balancing mass in plane M (l = 0.4 m, r = 0.5 m). Couples m r l : A −6.00, B 3.00, C 8.64, D 18.2 kg·m². $\Sigma H_c = -10.61$; $\Sigma V_c = -9.49$ ⇒ $|C| = 14.24$ ⇒ $m_M r_M = 14.24/0.4 = 35.6$ kg·m at 41.8° ⇒ m_M = 71.2 kg (r = 0.5). Force balance : $\Sigma H = 24.6$, $\Sigma V = 9.9$ kg·m ; adding $m_M r_M$ (26.5, 23.7) ⇒ $m_L r_L$ = 61.2 kg·m at 213.4° ⇒ m_L = 122.5 kg.

- **Remarks** : the choice of planes is limited by space ; the balancing masses are usually in the end planes of a rotor (turbine disc, armature) ; balancing machines detect the dynamic reaction at bearings. Residual unbalance is expressed as G (mm/s) grade, ISO 1940.
- **Balancing of a rotor in the field** : trial-weight method (two-plane), influence coefficients ; balancing of flexible rotors operating near critical speeds needs multi-plane balancing.
- (Reciprocating and multi-cylinder engine balancing – primary & secondary forces, in-line, V, radial – is outside this unit but follows the same force & couple method.)

5. Cams – Classification & Terminology

A cam is a rotating (or translating) machine element that gives a prescribed, usually non-uniform, motion to a follower by direct contact – a higher pair.



Classification of cams

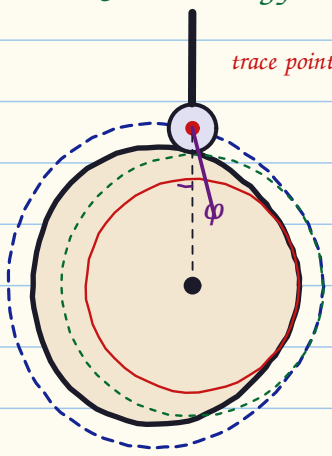
- **Radial (disc/plate) cams** – follower moves radially. **Translating (wedge)** – cam moves linearly. **Cylindrical (drum) cam** – follower moves parallel to the axis ; groove or end cam. Also conjugate, globoidal, spherical cams and cams with specified contour.

Classification of followers

Basis	Types	Remarks
Shape	Knife-edge / roller / flat-faced (mushroom) / spherical	knife-edge wears, roller common, flat for high speed
Motion	Reciprocating (translating) / oscillating (swinging arm)	valves / relay mechanisms
Line of motion	In-line (passes through cam centre) / offset	offset reduces pressure angle in rise
Closure	Spring, gravity, positive (groove, conjugate)	high speed – positive drive

Terminology

Cam terminology



Base circle (r_b) – smallest circle

Prime circle ($r_b + r_{roller}$)

Pitch curve – path of trace point

Cam profile – working surface

Pressure angle φ : normal ∧ follower path

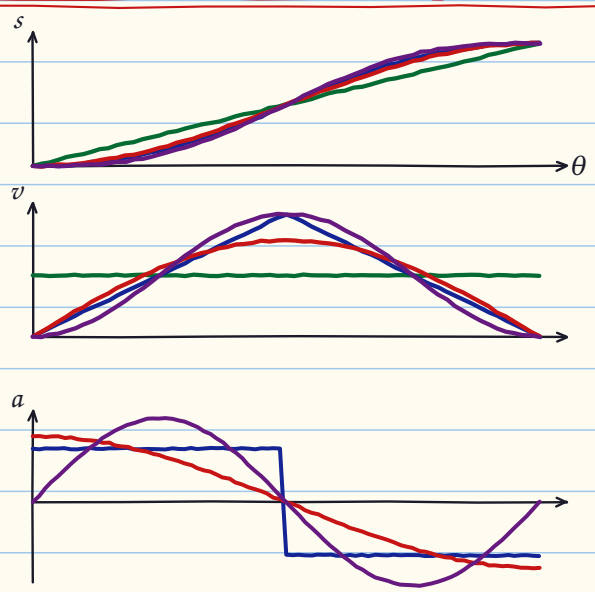
Lift (stroke) h = max s

Angles : rise δ_1 → dwell → return δ_2 → dwell

- **Base circle** : smallest circle tangent to the cam profile (radius r_b). **Trace point** : reference point on the follower (knife edge, roller centre). **Pitch curve** : path of the trace point relative to the cam. **Prime circle** : smallest circle from the cam centre tangent to the pitch curve (radius $R_p = r_b + r_{roller}$).
- **Pressure angle φ** : angle between the direction of follower motion and the common normal at the contact point. **Pitch point** : the point on the pitch curve with maximum pressure angle. **Lift (stroke) h** : maximum follower displacement.

- **Angle of ascent (rise) δ_1, dwell** (follower stationary), **angle of descent (return) δ_2,** remaining dwell. Total 360°. A displacement diagram plots s against θ (or time).

6. Follower Motions – Displacement Diagrams



Follower motions (rise h in δ)

Uniform velocity

$s = h\theta/\delta$; $v = h\omega/\delta$; $a = 0$ (∞ at ends)

Uniform acc./retardation

$s = 2h(\theta/\delta)^2$; $v_{max} = 2h\omega/\delta$; $a = \pm 4h\omega^2/\delta^2$

Simple harmonic

$s = (h/2)(1-\cos\pi\theta/\delta)$; $v_{max} = \pi h\omega/2\delta$; $a_{max} = \pi^2 h\omega^2/2\delta^2$

Cycloidal

$s = h[\theta/\delta - \sin(2\pi\theta/\delta)/2\pi]$; $v_{max} = 2h\omega/\delta$; $a_{max} = 2\pi h\omega^2/\delta^2$

Best : cycloidal (smooth a)

Cam rotates at ω ; rise h in cam angle δ (time $t = \delta/\omega$). Let θ = angle from start of rise.

(i) Uniform velocity

$$s = h\theta/\delta \quad v = h\omega/\delta \quad a = 0 \text{ (theoretically } \infty \text{ at the ends – infinite jerk/impact)}$$

- Straight-line displacement ; used only with smoothing arcs at the ends (low speed).

(ii) Uniform acceleration & retardation (parabolic)

$$0 \le \theta \le \delta/2 : s = 2h(\theta/\delta)^2 \text{ , } v = 4h\omega\theta/\delta^2 \text{ , } a = 4h\omega^2/\delta^2 \text{ (second half : retardation of equal magnitude)}$$

- Minimum possible peak acceleration for the given rise ($a_{max} = 4h\omega^2/\delta^2$, $v_{max} = 2h\omega/\delta$) but a sudden change of acceleration (finite jerk ∞ at start, middle, end) ; used for low-medium speeds.

(iii) Simple harmonic motion

$$s = (h/2)(1 - \cos(\pi\theta/\delta)) \quad v = (\pi h\omega/2\delta)\sin(\pi\theta/\delta) \quad a = (\pi^2 h\omega^2/2\delta^2)\cos(\pi\theta/\delta)$$

- $v_{max} = \pi h\omega/2\delta$ (mid-rise) ; $a_{max} = 4.93h\omega^2/\delta^2$ (start & end). Acceleration jumps from 0 to a_{max} at the ends of a dwell – step in a ⇒ moderate shock ; smooth if no

dwell (rise-return).

(iv) Cycloidal motion

$$s = h\,[\,\theta/\beta - \sin(2\pi\theta/\beta)/2\pi\,]\quad v = (h\omega/\beta)(1 - \cos(2\pi\theta/\beta))\quad a = (2\pi h\omega^2/\beta^2)\sin(2\pi\theta/\beta)$$

- $v_{max} = 2h\omega/\beta$; $a_{max} = 2\pi h\omega^2/\beta^2 = 6.28h\omega^2/\beta^2$. Acceleration is zero at the ends and varies sinusoidally – no sudden change of acceleration (finite jerk) ⇒ least vibration & wear ; best for high-speed cams.

Motion	v_{max}	a_{max}	Remark
Uniform velocity	$h\omega/\beta$	∞	impact at ends
Parabolic	$2h\omega/\beta$	$4h\omega^2/\beta^2$	lowest a_{max}
SHM	$1.57h\omega/\beta$	$4.93h\omega^2/\beta^2$	simple, a step at ends
Cycloidal	$2h\omega/\beta$	$6.28h\omega^2/\beta^2$	smoothest

Ex 3. $h = 40$ mm, rise angle $\beta = 60°$, $N = 600$ rpm ($\omega = 62.83$ rad/s, $\beta = 1.047$ rad) : uniform $v = 2.40$ m/s ; SHM $v_{max} = \pi(0.04)(62.83)/(2\times1.047) = 3.77$ m/s, $a_{max} = \pi^2(0.04)(62.83)^2/(2\times1.047^2) = 711$ m/s² ; cycloidal $v_{max} = 4.80$ m/s, $a_{max} = 905$ m/s² ; parabolic $a = 576$ m/s².

- **Derivatives of follower motion** : $s = f(\theta)$; $v = ds/dt = \omega\, ds/d\theta$; $a = \omega^2\, d^2s/d\theta^2$; jerk $j = \omega^3\, d^3s/d\theta^3$ (rate of change of acceleration – decides vibration & noise). Fall (return) motion is obtained by symmetry : same laws with θ measured from the start of return, $s = h - f(\theta)$.
- Choice : speed ↑ ⇒ move from uniform → SHM → cycloidal. Dwell-rise-dwell (DRD) ; dwell-rise-return-dwell (DRRD) ; dwell-rise-dwell-return (DRDR – most common).

7. Profile Construction & Specified-Contour Cams

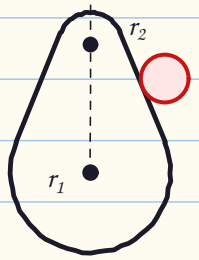
Graphical (inversion) method – steps

- (1) Draw the base circle, prime circle and the follower in its initial position. (2) Draw the displacement diagram with the same rise/dwell angles ; divide into equal parts. (3) Divide the cam circle in the same angular intervals measured opposite to cam rotation (kinematic inversion – cam fixed, follower rotating). (4) Along each radial line, mark the follower displacement from the base/prime circle.

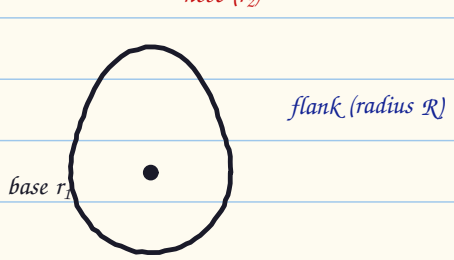
(5) Join the points – pitch curve. (6) For a roller follower, draw the roller at each point and take the envelope – cam profile (for flat-faced – draw the face line, envelope). Offset follower : lines tangent to the offset circle.

Cams with specified contours

Tangent cam (roller follower)



Circular-arc cam



flank : $s = (r_1+r_r)(\sec\theta - 1)$

nose : $x = OQ\cos\theta + \sqrt{((r_2+r_r)^2 - OQ^2\sin^2\theta)}$

- ***Tangent cam*** : the flanks are straight lines tangent to the base circle (r_1) and nose circle (r_2). For a roller follower with roller radius r_r and in-line motion, while the roller touches the straight flank the displacement from the base position :

$$s = (r_1 + r_r)(\sec\theta - 1) \quad v = (r_1 + r_r)\,\omega\,\sec\theta\tan\theta \quad a = (r_1 + r_r)\,\omega^2\,\sec\theta\,(\sec^2\theta + \tan^2\theta)$$

where θ = cam angle measured from the position when the roller starts leaving the base circle (the line of centres is perpendicular to the flank). When the roller reaches the nose circle (radius r_2) :

$$x = OQ\cos\theta + \sqrt{(r_2 + r_r)^2 - OQ^2\sin^2\theta} \quad (OQ = \text{distance between cam centre and nose centre})$$

Ex 4. Tangent cam $r_1 = 30$, $r_r = 15$ mm ($\mathcal{R}_p = 45$ mm), $N = 300$ rpm ; roller on flank at $\theta = 20°$: $s = 45(\sec 20° - 1) = 2.89$ mm ; $v = 45 \times 31.42 \times 1.064 \times 0.364 = 0.548$ m/s ; at $\theta = 30°$ $s = 6.96$ mm.

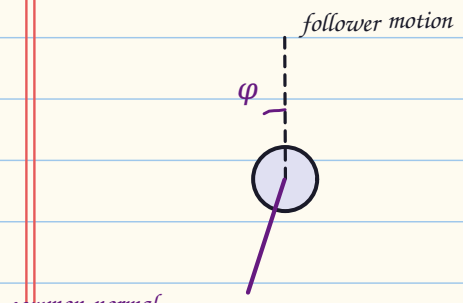
- ***Circular-arc (convex) cam*** : flanks are circular arcs of radius $\mathcal{R}$ (tangent to base circle & nose circle) – the cam has base circle r_1, nose r_2. While the roller is on the nose : $s = x - (r_1 + r_r)$ with $x = OQ\cos\theta + \sqrt{((r_2+r_r)^2 - OQ^2\sin^2\theta)}$, $OQ = r_1 + h - r_2$. On the flank the same formula with flank centre and radius $(\mathcal{R} + r_r)$.

Ex 5. Circular-arc nose : $r_1 = 30$, $h = 20$, $r_2 = 10$, $r_r = 15$ mm $\Rightarrow OQ = 40$ mm. At $\theta = 0$: $x = 40 + 25 = 65 \Rightarrow s = 20$ mm □ ; at $\theta = 20°$: $x = 40\cos 20° + \sqrt{(25^2 - 40^2\sin^2 20°)} = 37.6 + 20.9 = 58.5 \Rightarrow s = 13.5$ mm.

- Both cams are easy to manufacture (turned arcs and flats), used in automobile valve gear, but acceleration is not controlled (tangent cams give large acceleration at the nose). **Flat-faced follower** : pressure angle is zero ; contact point moves along the face by $(ds/d\theta)$ from the follower axis, so the face half-width must exceed $(ds/d\theta)_{max}$; cam size is limited by $\mathcal{R}_b + s + d^2s/d\theta^2 > 0$ (positive radius of curvature).

8. Pressure Angle & Undercutting

Pressure angle



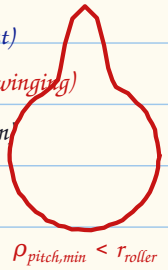
$\tan\varphi = (ds/d\theta) / (\mathcal{R}_p + s)$ (in-line)

$\tan\varphi = (ds/d\theta \mp e)/(\sqrt{(\mathcal{R}_p^2 - e^2)} + s)$ (offset)

Limits : $\varphi_{max} \approx 30°$ (translating), 45° (swinging)

↑ prime radius $\mathcal{R}_p \Rightarrow \downarrow \varphi$ (but bigger cam)

Undercutting (cusp / loop)



- **Pressure angle** φ – the angle between the common normal and the follower path. The component of the contact force along the follower = $\mathcal{F}\cos\varphi$ (useful), perpendicular component $\mathcal{F}\sin\varphi$ presses the follower sideways against its guide ⇒ friction, jamming, bending when φ is large.

$$\tan\varphi = (ds/d\theta) / (\mathcal{R}_p + s) \text{ (in-line follower, } ds/d\theta = v/\omega)$$

- Limiting values : $\varphi_{max} \approx 30°$ (reciprocating follower), 45° (oscillating). φ is largest near the middle of the rise for SHM.
- Reduction of φ : larger prime/base circle, offset the follower suitably, increase the cam rise angle, choose a favourable motion, use oscillating follower.

Ex 6. SHM rise $h = 20$ mm, $\delta = 60°$, $\mathcal{R}_p = 40$ mm : $ds/d\theta = (\pi h/2\delta)\sin(\pi\theta/\delta) = 30\sin\chi$ mm/rad ; at $\chi \approx 1.4$ rad (≈ 80° of the rise) : $s = 10(1 - \cos 1.4) = 8.3 \Rightarrow \tan\varphi = 29.5/(40 + 8.3) = 0.61 \Rightarrow \varphi = 31.4°$ (> 30° □). Increase $\mathcal{R}_p$ to 50 mm : $\tan\varphi = 29.5/58.3 = 0.506 \Rightarrow \varphi = 26.8°$ □.

Undercutting

- **Undercutting** (cusping) : the pitch curve has a radius of curvature smaller than the roller radius (convex portion), so the roller cannot trace it – the profile

develops a loop/cusp and the follower motion is altered.

$$\rho_{pitch} = [\, \mathcal{R}^2 + (d\mathcal{R}/d\theta)^2 \,]^{3/2} / [\, \mathcal{R}^2 + 2(d\mathcal{R}/d\theta)^2 - \mathcal{R}\, d^2\mathcal{R}/d\theta^2 \,] > r_r \quad (\mathcal{R} = \mathcal{R}_p + s)$$

- Avoid by : smaller roller ($r_r < 0.8\,\rho_{min}$), larger base circle, gentler motion (reduce a), limit the cam rise per angle, use a flat-faced follower only with $\mathcal{R}_b + s + d^2s/d\theta^2 > 0$.

Tabulated displacements ($h = 40$ mm, $\delta = 60°$)

θ (deg)	0	10	20	30	40	50	60
SHM s (mm)	0	2.68	10.0	20.0	30.0	37.3	40
Cycloidal s (mm)	0	1.16	7.82	20.0	32.2	38.8	40
Parabolic s (mm)	0	2.22	8.89	20.0	31.1	37.8	40

Note : cycloidal starts slowest (zero acceleration at the ends) ; parabolic has a sudden change of acceleration at the mid-point ($\theta = 30°$).

Design hints for cam–follower

- Choose the follower type by speed and load (flat or roller). Choose motion (cycloidal for high speed). Find the smallest prime circle that keeps $\varphi_{max} \le 30°$; check undercutting on the pitch curve ; offset the follower to reduce φ on the rise stroke. Specify tolerance and hardness (case-hardened, ground profile) because contact stresses are Hertzian.
- Jump phenomenon : at high speed the follower loses contact when the inertia force exceeds the spring force – spring force must exceed $m\,a_{max}$ + weight ⇒ spring stiffness $k \ge (m\,a_{max})/(\text{preload} + h)$.

10. Practical Balancing Notes

- **Balancing machines** : stand-alone (soft-bearing) and hard-bearing types measure the bearing reaction due to unbalance, then give the magnitude and angular position of correction mass in two planes.
- Correction by drilling (removing metal), adding weights, mass-adjusting screws ; for fans — trim weights on blades.

- Allowed residual unbalance $e_{per}\cdot\omega = G$ (mm/s) – for rigid rotors G = 6.3 (fans, pumps, motors), 2.5 (turbines, compressors), 1 (grinders). Example: a 12 kg rotor at 3000 rpm (ω = 314 rad/s), G = 2.5 : e_{per} = 2.5/314 = 0.008 mm ⇒ permissible unbalance U = 12 × 0.008 = 96 g·mm.

11. Additional Solved Problems

P1 (static balance). Two masses 3 kg at 0.2 m (0°) and 4 kg at 0.15 m (90°) in one plane : m r = 0.6 and 0.6 ⇒ $R = \sqrt{(0.36 + 0.36)} = 0.849$ kg·m at 45° ⇒ balancing mass 0.849 kg·m at 225° (e.g. 4.24 kg at 0.2 m).

P2 (two-plane). A mass of 10 kg at r = 0.2 m (m r = 2 kg·m) lies between planes P and Q, 0.3 m from P and 0.5 m from Q. $m_P r_P = 2 \times 0.5/0.8 = 1.25$ kg·m ; $m_Q r_Q = 2 \times 0.3/0.8 = 0.75$ kg·m ; sum 2.0 □ (both opposite to the 10 kg mass).

P3 (cam). SHM follower h = 30 mm rise in 90°, N = 120 rpm : $v_{max} = \pi h\omega/2\beta = \pi(0.03)(12.57)/(2\times1.571) = 0.377$ m/s ; $a_{max} = \pi^2 h\omega^2/2\beta^2 = 9.87(0.03)(157.9)/(2\times2.467) = 9.47$ m/s².

P4 (cycloidal). Same data : $a_{max} = 2\pi h\omega^2/\beta^2 = 6.283(0.03)(157.9)/2.467 = 12.1$ m/s².

Short answers – Quick Revision

- Why is a statically balanced rotor not necessarily dynamically balanced? The couple of the forces in different planes remains (Σ m r l ≠ 0).
- Why use cycloidal motion for high-speed cams? Acceleration starts and ends at zero – low jerk, vibration and wear.
- Effect of increasing the base circle? Smaller pressure angle, less undercutting – larger cam size.
- Which follower gives zero pressure angle? Flat-faced follower with flat perpendicular to its motion.
- Difference between pitch curve and cam profile? Pitch curve is the locus of the trace point ; the profile is offset inward by the roller radius.

12. Quick Formula Sheet – Unit IV

Item	Formula
Centrifugal force	$F = m\omega^2 r$

Item	Formula
Balance (one plane)	$\Sigma\ m\ r = 0\ ;\ m_b r_b = \mathcal{R}$
Balance (many planes)	$\Sigma\ m\ r = 0\ ;\ \Sigma\ m\ r\ l = 0$
SHM	$s = (h/2)(1-\cos\pi\theta/\beta)\ ;\ a_{max} = \pi^2 h\omega^2/2\beta^2$
Cycloidal	$a_{max} = 2\pi h\omega^2/\beta^2\ ;\ v_{max} = 2h\omega/\beta$
Parabolic	$a = 4h\omega^2/\beta^2\ ;\ v_{max} = 2h\omega/\beta$
Tangent cam	$s = (r_1+r_r)(\sec\theta - 1)$
Pressure angle	$\tan\varphi = (ds/d\theta)/(\mathcal{R}_p+s)$
Undercutting	$\rho_{pitch,min} > r_{roller}$

13. Important Questions

- Distinguish static and dynamic balancing. Explain the balancing of several masses rotating in different planes (analytical & graphical).
- Classify cams and followers with sketches. Define base circle, prime circle, pitch curve, pressure angle.
- Derive displacement, velocity and acceleration for SHM, uniform acceleration & cycloidal motion; compare them.
- Explain the construction of a cam profile for a roller follower ; describe tangent and circular-arc cams with expressions.
- What is pressure angle? How does it vary with base circle? What is undercutting and how is it avoided?

UNIT – V

Vibrations, Turning Moment & Flywheels

Subject : Theory of Machines (23A03403)

Syllabus: Vibrations – introduction, DOF, types, free natural vibration (Newton & energy methods), damped (under, critical, over), forced vibration with and without damping, vibration isolation & transmissibility. Turning moment diagrams (steam engine, I.C. engine, multi-cylinder), crank effort, coefficients of fluctuation of energy & speed, flywheel design, flywheel for punching press.

1. Vibrations – Introduction

Vibration = repetitive (oscillatory) motion of a body about an equilibrium position. Causes : unbalance, elastic forces, external excitation. Effects : noise, fatigue, wear, discomfort, resonance failure ; useful in vibrators, musical instruments, ultrasonic cleaners.

- Terms : **period** T (time of one cycle), **frequency** $f = 1/T$ (Hz), **circular frequency** $\omega = 2\pi f$ (rad/s), **amplitude** X (maximum displacement), **phase** φ, **natural frequency** ω_n (frequency of free vibration), **resonance** (forcing frequency ≈ natural frequency).
- **Simple harmonic motion** : $x = X \sin(\omega t + \varphi)$; velocity $\dot{x} = X\omega \cos(\omega t + \varphi)$; acceleration $\ddot{x} = -\omega^2 x$ (proportional to displacement and directed to the mean position).

Degrees of freedom

Minimum number of independent coordinates required to define the motion : spring–mass (1), simple pendulum (1), double pendulum (2), two-rotor torsional system (2), a beam/continuous system (∞).

Types of vibration

Basis	Types	Remark
External force	Free (only initial disturbance) / Forced (periodic force)	ω_n / forcing ω

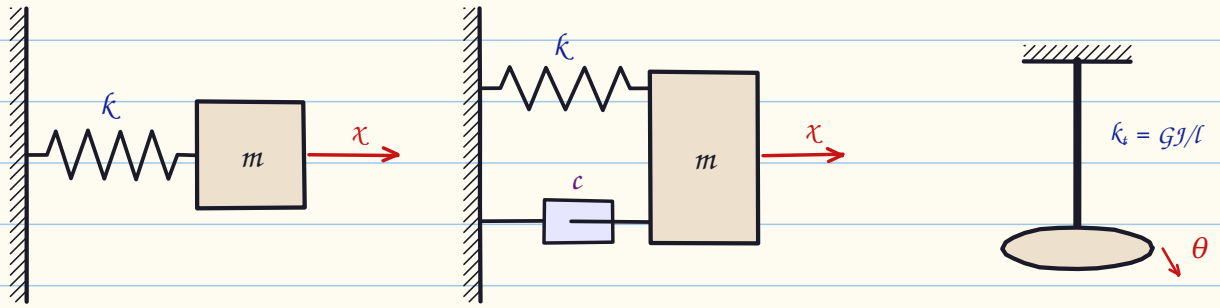
Basis	Types	Remark
Damping	Undamped / damped (viscous, Coulomb, structural)	energy dissipated
Motion	Longitudinal / transverse / torsional	spring-mass / beam / shaft
System	Linear / non-linear ; discrete / continuous	superposition holds for linear
Excitation	Deterministic / random	earthquake, road roughness

Equivalent stiffness

Case	Stiffness	Remark
Springs in parallel	$k = k_1 + k_2$	same deflection
Springs in series	$1/k = 1/k_1 + 1/k_2$	same force
Cantilever (end load)	$k = 3EI/l^3$	ω_n from m
Simply-supported beam, central load	$k = 48EI/l^3$	
Fixed–fixed beam, central load	$k = 192EI/l^3$	
Shaft (torsional)	$k_t = GJ/l = \pi G d^4/(32\,l)$	N·m/rad

2. Free Undamped Vibration (SDOF)

(a) Spring–mass (b) Spring–mass–damper (c) Torsional



$m\ddot{x} + kx = 0$ $m\ddot{x} + c\dot{x} + kx = F(t)$ $I\ddot{\theta} + k_t\theta = 0$

Newton's (equilibrium) method

Mass m on a spring of stiffness k, displacement x from the static equilibrium (weight is balanced by the static spring force, so it cancels) :

$$m\ddot{x} + kx = 0 \Rightarrow \omega_n = \sqrt{(k/m)}\ f_n = (1/2\pi)\sqrt{(k/m)}\ T = 2\pi\sqrt{(m/k)}$$

- Solution : $x = A\cos\omega_n t + B\sin\omega_n t = X\sin(\omega_n t + \varphi)$; $A = x_0$, $B = \dot{x}_0/\omega_n$; $X = \sqrt{(x_0^2 + (\dot{x}_0/\omega_n)^2)}$.
- **Static deflection** $\delta_{st} = mg/k \Rightarrow \omega_n = \sqrt{(g/\delta_{st})}$; $f_n = 0.4985/\sqrt{\delta_{st}}$ (δ_{st} in m).
- **Torsional** : $I\ddot{\theta} + k_t\theta = 0 \Rightarrow \omega_n = \sqrt{(k_t/I)}$; **simple pendulum** $\omega_n = \sqrt{(g/l)}$; **compound pendulum** $\omega_n = \sqrt{(g\,h/(k_G^2 + h^2))}$.
- **Two rotors on one shaft** (two-DOF) : node at distance $l_1 = l\,I_2/(I_1 + I_2)$ from rotor 1 ; $\omega_n = \sqrt{(\,k_t(I_1 + I_2)/(I_1\,I_2)\,)}$, $k_t = GJ/l$.

Energy (Rayleigh) method

Conservative system : total energy constant $\Rightarrow KE_{max} = PE_{max}$ (or $d(KE + PE)/dt = 0$).

$$\tfrac{1}{2}\,m\,\dot{x}_{max}^2 = \tfrac{1}{2}\,k\,x_{max}^2 \text{ with } \dot{x}_{max} = \omega_n x_{max} \Rightarrow \omega_n^2 = k/m$$

- **Effect of spring mass** m_s : effective mass $m + m_s/3 \Rightarrow \omega_n = \sqrt{(k/(m + m_s/3))}$. Pulley/rolling cylinder systems : include rotational KE.

Ex 1. $m = 5$ kg, $k = 2000$ N/m : $\omega_n = \sqrt{(2000/5)} = 20$ rad/s ; $f_n = 3.18$ Hz ; $T = 0.314$ s ; $\delta_{st} = 5\times9.81/2000 = 24.5$ mm. Check : $0.4985/\sqrt{0.0245} = 3.18$ Hz ☐.

Ex 2 (energy method). Uniform cylinder $m = 10$ kg, $R = 0.2$ m rolls without slipping, axle attached to spring $k = 3000$ N/m : $KE = \tfrac{1}{2}m\dot{x}^2 + \tfrac{1}{2}(\tfrac{1}{2}mR^2)(\dot{x}/R)^2 = \tfrac{3}{4}\,m\,\dot{x}^2 \Rightarrow m_{eq} = 1.5\,m = 15$ kg $\Rightarrow \omega_n = \sqrt{(3000/15)} = 14.1$ rad/s.

Ex 3 (springs). Two springs 4000 and 6000 N/m : in parallel $k = 10\,000$ N/m ; in series $k = 2400$ N/m. A 10 kg mass on the series pair : $\omega_n = \sqrt{(2400/10)} = 15.5$ rad/s.

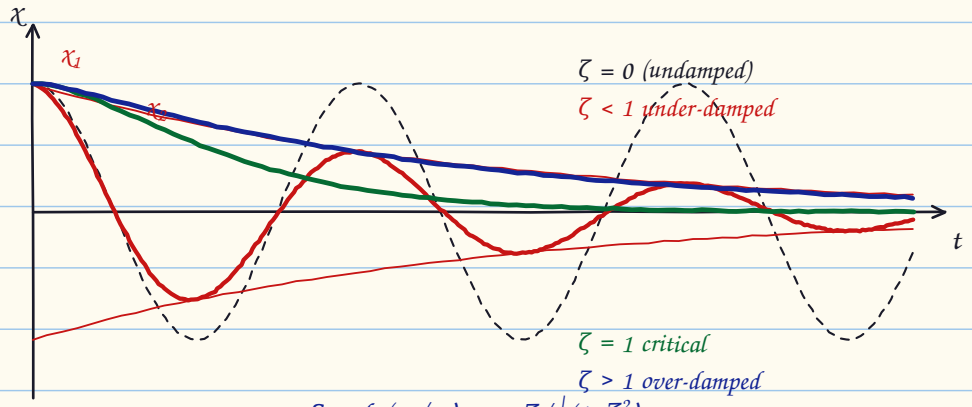
3. Free Damped Vibration

Viscous damping force $F_d = c\,\dot{x}$ (c in N·s/m) :

$$m\ddot{x} + c\dot{x} + kx = 0 \quad \zeta = c/c_c\ ,\ c_c = 2\sqrt{(km)} = 2m\omega_n \text{ (critical damping)}$$

Characteristic equation $s^2 + 2\zeta\omega_n s + \omega_n^2 = 0 \Rightarrow s = -\zeta\omega_n \pm \omega_n\sqrt{(\zeta^2 - 1)}$.

Free vibration with viscous damping



$\delta = \ln(x_1/x_2) = 2\pi\zeta/\sqrt{(1-\zeta^2)}$

Case	Solution	Motion
Under-damped $\zeta < 1$	$x = X\, e^{-\zeta\omega_n t}\} \sin(\omega_d t + \varphi)$, $\omega_d = \omega_n\sqrt{(1-\zeta^2)}$	oscillatory, decaying amplitude
Critically damped $\zeta = 1$	$x = (A + Bt)\, e^{-\omega_n t}\}$	no oscillation, fastest return to rest
Over-damped $\zeta > 1$	$x = A\, e^{s_1 t}\} + B\, e^{s_2 t}\}$ ($s_{1,2}$ real negative)	aperiodic, sluggish return

- Under-damped : successive peaks decrease in geometric progression. **Logarithmic decrement** (ratio of two successive amplitudes, period $T_d = 2\pi/\omega_d$) :

$$\delta = \ln(x_1/x_2) = \zeta\omega_n T_d = 2\pi\zeta/\sqrt{(1-\zeta^2)} \approx 2\pi\zeta \text{ (small } \zeta) \quad \delta = (1/n)\ln(x_0/x_n) \text{ for } n \text{ cycles}$$

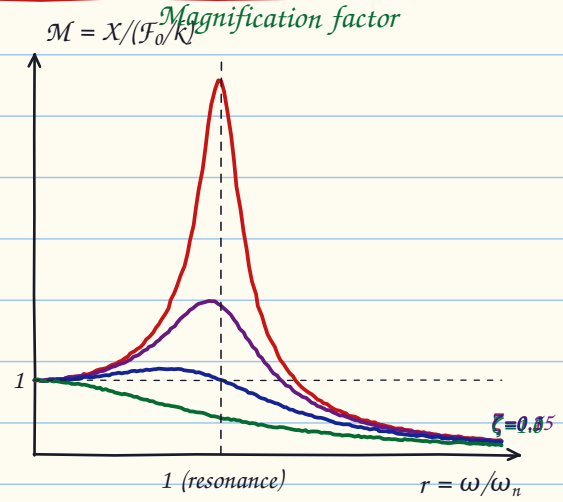
- Use δ to measure ζ experimentally : $\zeta = \delta/\sqrt{(4\pi^2 + \delta^2)}$.
- Damped frequency $\omega_d < \omega_n$ (very little difference for $\zeta < 0.2$). Time constant $\tau = 1/(\zeta\omega_n)$.
- **Coulomb (dry-friction) damping** : amplitude decreases linearly, by $4\mu N/k$ per cycle ; motion stops when $kx \le \mu N$. Structural (hysteretic) damping $\propto$ displacement.
- Practical use of critical damping : door closers, galvanometers, gun recoil, suspension ($\zeta \approx 0.2$–0.4 in cars), instrument pointers.

Ex 4. $m = 10$ kg, $k = 4000$ N/m, $c = 40$ N·s/m : $\omega_n = 20$ rad/s ; $c_c = 2\sqrt{(4000\times10)} = 400$; $\zeta = 0.1$; $\omega_d = 20\sqrt{0.99} = 19.9$ rad/s ; $\delta = 2\pi(0.1)/\sqrt{0.99} = 0.631 \Rightarrow x_1/x_2 = e^{0.631} = 1.88$. Amplitude after 5 cycles $= x_0/1.88^5 = 4.3$ % of x_0.

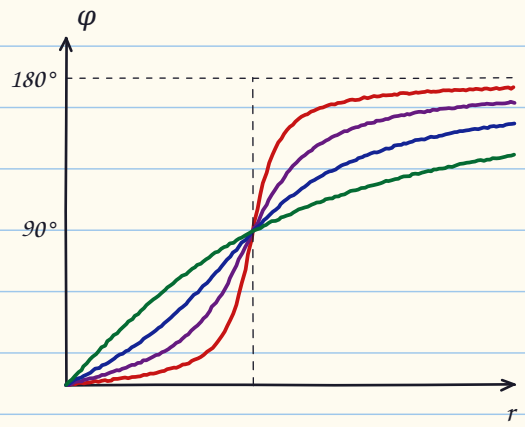
Ex 5. Observed amplitude ratio over 10 cycles is $2:1 \Rightarrow \delta = (1/10)\ln 2 = 0.0693$ $\Rightarrow \zeta = 0.0693/\sqrt{(39.5 + 0.0048)} = 0.011$.

4. Forced Vibration

Magnification factor



Phase angle



(a) Without damping

Harmonic force $F_0 \sin\omega t$: $m\ddot{x} + kx = F_0 \sin\omega t$. Steady-state amplitude :

$$X = (F_0/k) / (1 - r^2)\ r = \omega/\omega_n$$

- For $r < 1$ the motion is in phase with the force ; for $r > 1$ it is 180° out of phase (sign negative). At $r = 1$: resonance – amplitude theoretically infinite (X grows linearly with time).

(b) With viscous damping

$m\ddot{x} + c\dot{x} + kx = F_0 \sin\omega t \Rightarrow$ steady-state $x = X \sin(\omega t - \varphi)$:

$$X = (F_0/k) / \sqrt{(\ (1 - r^2)^2 + (2\zeta r)^2\ )}\ \tan\varphi = 2\zeta r / (1 - r^2)$$

- **Magnification factor** $M = X/(F_0/k) = 1/\sqrt{((1 - r^2)^2 + (2\zeta r)^2)}$. $M \to 1$ for $r \to 0$ (static) ; $M \to 0$ for $r \gg 1$ (mass cannot follow). At resonance ($r = 1$) : $M = 1/(2\zeta)$ and $\varphi = 90°$ irrespective of damping. Peak occurs at $r = \sqrt{(1 - 2\zeta^2)}$: $M_{max} = 1/(2\zeta\sqrt{(1 - \zeta^2)})$.
- Damping matters only near resonance ; far from resonance the system is mass or stiffness controlled. The total response = transient (dies out) + steady-state. Q-factor $Q = 1/(2\zeta)$; half-power bandwidth $\Delta\omega = 2\zeta\omega_n$.

Ex 6. $m = 20$ kg, $k = 8000$ N/m, $c = 80$ N·s/m, $F_0 = 100$ N, $\omega = 15$ rad/s : $\omega_n = 20$; $r = 0.75$; $c_c = 800 \Rightarrow \zeta = 0.1$. $X = (100/8000)/\sqrt{((1 - 0.5625)^2 + (0.15)^2)} = 0.0125/0.4625 =$ 27.0 mm ; $\varphi = \tan^{-1}(0.15/0.4375) =$ 18.9°. At resonance $X = 0.0125/0.2 = 62.5$ mm.

(c) Rotating unbalance (e.g. unbalanced motor)

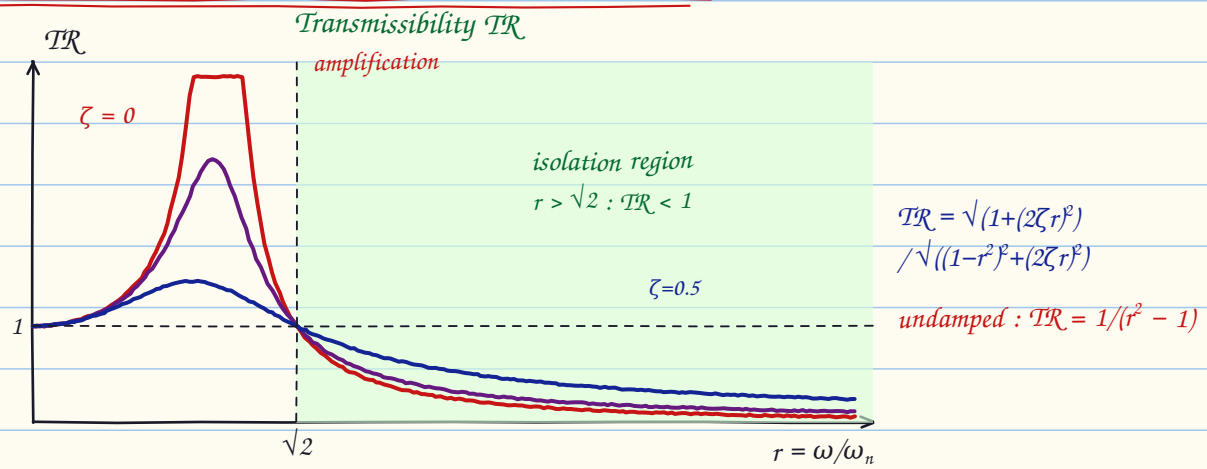
Total mass M, unbalanced mass m_u at eccentricity e, speed ω : forcing amplitude $F_0 = m_u\, e\, \omega^2$.

$$X = (m_u\, e/M)\, r^2 / \sqrt{(1 - r^2)^2 + (2\zeta r)^2}\ \ X \to m_u e/M \ (r \gg 1)$$

(d) Support (base) excitation & whirling of shafts

- Base moves $y = Y \sin\omega t$: absolute response ratio $X/Y = \sqrt{(1 + (2\zeta r)^2)} / \sqrt{((1 - r^2)^2 + (2\zeta r)^2)}$ = transmissibility. Vehicle suspensions, seismographs ($r \gg 1$), accelerometers ($r \ll 1$).
- **Critical speed of a shaft** (whirling) $N_c = 30/\pi \times \sqrt{(g/\delta_{st})}$ = natural frequency of transverse vibration ; operate well above or below it.

5. Vibration Isolation & Transmissibility



Isolation reduces the force transmitted from a machine to the foundation (force isolation) or the motion transmitted from the foundation to the equipment (motion isolation).

- Force transmitted $F_T = \sqrt{((kX)^2 + (c\omega X)^2)} \Rightarrow$ **transmissibility ratio** :

$$TR = F_T/F_0 = \sqrt{(1 + (2\zeta r)^2)} / \sqrt{(1 - r^2)^2 + (2\zeta r)^2}\ \ (\text{undamped} : TR = 1/(r^2 - 1))$$

- $TR < 1$ only for $r > \sqrt{2}$. So the natural frequency of the mount must be much lower than the disturbing frequency : $f_n < f/\sqrt{2}$. Isolation efficiency $\eta = (1 - TR) \times 100$ %.
- **Effect of damping** : increases isolation near resonance ($r \approx 1$) but reduces it for $r > \sqrt{2}$ (TR becomes larger) – a compromise : $\zeta \approx 0.05$–0.2 for passing through

resonance during start-up.

- Methods : steel springs, rubber pads, cork, felt, pneumatic mounts, inertia blocks (increase m), dynamic vibration absorber (tuned mass damper) to cancel a given forcing frequency.

Ex 7. Machine 100 kg, 1200 rpm (ω = 125.7 rad/s) ; required TR = 0.25 (efficiency 75 %), undamped isolators : $r^2 = 1 + 1/TR = 5 \Rightarrow r = 2.236 \Rightarrow \omega_n$ = 56.2 rad/s $\Rightarrow k = m\omega_n^2$ = 316 kN/m ; static deflection $\delta_{st} = g/\omega_n^2$ = 3.1 mm ; transmitted force = 0.25 F_0.

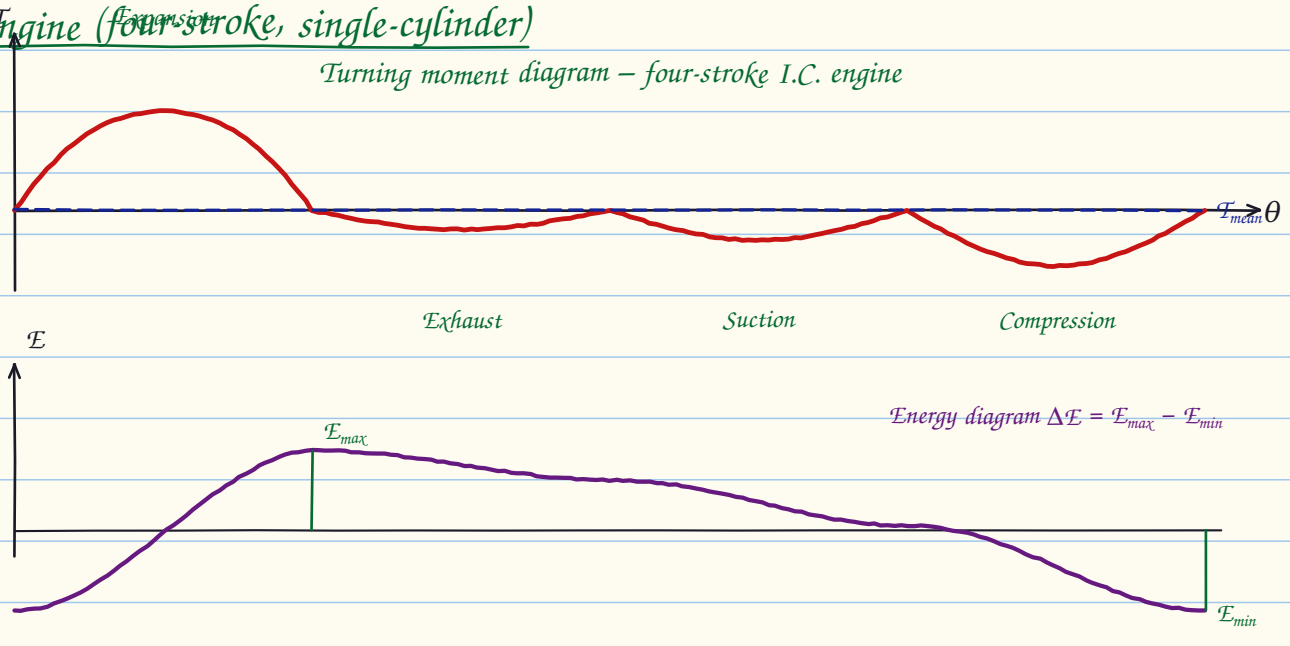
Ex 8. Amplitude of transmitted force with ζ = 0.1 at r = 3 : $TR = \sqrt{(1 + 0.36)}/\sqrt{(64 + 0.36)} = 1.166/8.02 = 0.145$ (vs 0.125 undamped) – damping slightly worse above $\sqrt{2}$.

6. Turning Moment Diagrams

Turning moment (crank effort) $T = F_p\, r \sin(\theta + \varphi)/\cos\varphi$ varies with crank angle because the piston force varies. The turning-moment diagram plots T against θ for one cycle.

- **Mean (average) torque** T_{mean} = (work done per cycle)/(angle of cycle) : 2π for steam engine / two-stroke, 4π for four-stroke ; $P = T_{mean}\,\omega$. The area under the curve over the cycle = work per cycle.

I.C. Engine (four-stroke, single-cylinder)



- Cycle = 720° : expansion (positive large torque), exhaust, suction, compression (negative torque – work done on the gas). Only one stroke in four does positive

work ⇒ very large fluctuations ⇒ heavy flywheel.

- **Steam engine (double-acting)** : torque produced in both strokes ; the diagram has two humps per revolution (θ = 0–180° and 180°–360°), never negative ⇒ small fluctuation.
- **Multi-cylinder engines** : individual curves, shifted by the phase angle (720°/number of cylinders for four-stroke), are added ⇒ smoother resultant, smaller flywheel (fluctuation ratio falls from ≈ 2 for single-cylinder four-stroke to ≈ 0.2 for four-cylinder).
- **Crank effort** = torque on the crankshaft = T (above) ; reaction torque on frame equal and opposite.

Fluctuation of energy & speed

- Above T_{mean} the engine produces more energy than the load needs ⇒ flywheel speeds up ; below T_{mean} it gives energy back ⇒ slows down.

$\Delta E = E_{max} - E_{min}$ (maximum fluctuation of energy = difference between the highest and lowest points of the cumulative-energy diagram)

- **Coefficient of fluctuation of energy** $C_E = \Delta E$ / (work done per cycle). **Coefficient of fluctuation of speed** $C_s = (\omega_1 - \omega_2)/\omega_{mean} = (N_1 - N_2)/N_{mean}$, $\omega_{mean} = (\omega_1 + \omega_2)/2$. Coefficient of steadiness $m = 1/C_s$.

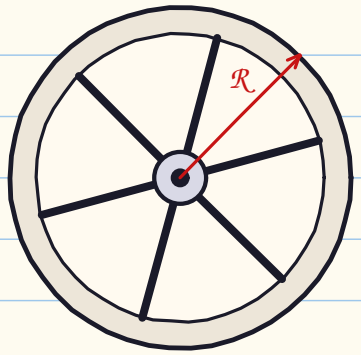
Machine	C_s
Crushing machines	0.2
Punching, shearing	0.03 – 0.05
Pumps, machine tools	0.03 – 0.05
Weaving / spinning	0.01 – 0.02
Automobile engines	0.02 – 0.05
Generators, turbines	0.002 – 0.01
Alternators / dc generators for lighting	0.0035 – 0.005

7. Flywheel

A heavy rotating wheel (rim + arms + hub) mounted on the crankshaft that stores kinetic energy when the engine produces excess energy and releases it when the engine produces less, keeping the speed within limits. (Differs from a governor : flywheel controls the **cyclic** speed variation within one cycle ; governor controls the **mean** speed

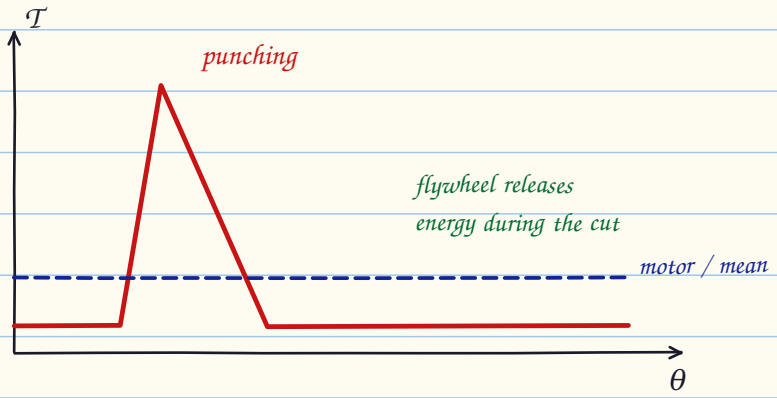
over many cycles with load changes.)

Flywheel (rim, arms, hub)



$\Delta E = I\,\omega^2\,C_s = m\,k^2\,\omega^2\,C_s$

Punching-press torque demand



$E_{punch} - E_{motor\ in\ punching\ time} = \Delta E$

$$\Delta E = I\,(\omega_1^2 - \omega_2^2)/2 = I\,\omega^2\,C_s = m\,k^2\,\omega^2\,C_s \approx m\,v^2\,C_s \;(\text{rim},\ k \approx R,\ v = \omega R)$$

Design of the flywheel

- (1) Find ΔE from the turning-moment (energy) diagram ; (2) choose C_s ; (3) $I = \Delta E/(\omega^2 C_s)$; (4) assume mean rim radius R (or v) and find rim mass $m = I/R^2$ (neglect arms and hub, ≈ 90 % of I in the rim) ; (5) rim cross-section $A = b\,t = m/(\rho\,2\pi R)$ ($b = (1–2)\,t$) ; (6) check hoop stress.
- **Hoop (centrifugal) stress** in the rim :

$$\sigma = \rho\,v^2 \Rightarrow v_{max} = \sqrt{(\sigma_{allow}/\rho)}$$

Cast iron $\rho \approx 7200\ kg/m^3$, $\sigma_{allow} \approx 6\ MPa \Rightarrow v \approx 29\ m/s$; cast steel $\rho = 7800$, $\sigma \approx 20–40\ MPa \Rightarrow v \approx 50–70\ m/s$.

- Arms (4–8 elliptical spokes) carry the torque (bending stress $\sigma_b = T/(Z)$) ; hub keyed to shaft ; large flywheels are split / segmental. Flywheels of high-speed energy storage use composite rims.

Ex 9. Energy areas of one cycle (J) about T_{mean} : +52, −124, +92, −140, +85, +35. Cumulative : E_0+52 → E_0−72 → E_0+20 → E_0−120 → E_0−35 → E_0. $E_{max} = E_0$+52 ; $E_{min} = E_0$−120 ⇒ ΔE = 172 J. N = 600 rpm (ω = 62.83), C_s = 0.02 : $I = 172/(62.83^2 \times 0.02)$ = 2.18 kg·m² ; for rim R = 0.3 m : m = 2.18/0.09 = 24.2 kg.

Ex 10. Rim, cast iron : R = 0.6 m, N = 300 rpm ⇒ v = 18.85 m/s ⇒ $\sigma = 7250 \times 18.85^2$ = 2.58 MPa (safe, below 6 MPa).

Ex 11. 4-stroke engine, 20 kW at 300 rpm : $T_{mean} = 20000/31.42 = 637$ N·m ; work per cycle = $T_{mean} \times 4\pi$ = 8000 J (= 20 kW × 0.4 s). If ΔE = 1.0 × work per cycle (illustrative) and $C_s = 0.02$: $I = 8000/(31.42^2 \times 0.02) = 405$ kg·m² ; maximum / minimum speeds $N_{1,2} = 300(1 \pm 0.01) = 303 / 297$ rpm.

8. Flywheel for Punching Press

Punching, shearing, riveting presses need a large energy during a very short part of the cycle but little in the rest ⇒ the motor is sized for the average power and the flywheel supplies the peak.

- Work done per punch (assuming force falls linearly from F_{max} to 0 during the plate thickness t) : $E_{punch} = \tfrac{1}{2} F_{max}\, t$, $F_{max} = \pi\, d\, t\, \tau_u$ (circular hole dia d in plate of thickness t, ultimate shear strength τ_u).
- If the cycle time is t_c (time per punch) and the punching (actual cutting) time is t_p : energy supplied by the motor during the punch = $(E_{punch}/t_c) \times t_p$; **energy to be supplied by the flywheel** :

$$\Delta E = E_{punch} - E_{punch}\, t_p/t_c = E_{punch}\,(1 - t_p/t_c)$$

- Then $I = \Delta E/(\omega^2 C_s)$; for efficiency η_m the motor power = $E_{punch}/(t_c\, \eta_m)$. The maximum speed fall during the punch : $\omega_2 = \omega(1 - C_s/2)$. The flywheel regains its speed during the remainder of the cycle.

Ex 12. d = 20 mm hole in 15 mm plate, τ_u = 350 MPa : $F_{max} = \pi(0.02)(0.015)(350\times10^6) = 330$ kN ; $E_{punch} = \tfrac{1}{2} \times 330\,000 \times 0.015 = 2474$ J. 30 holes/min (t_c = 2 s), punching time t_p = 0.4 s ⇒ motor supplies 2474 × 0.4/2 = 495 J ⇒ $\Delta E = 1979$ J. N = 300 rpm (31.42 rad/s), $C_s = 0.02$: $I = 1979/(31.42^2 \times 0.02) = 100$ kg·m² ; mean motor power = 2474/2 = 1237 W (≈ 1.24 kW).

- Concept : a lighter flywheel needs a bigger motor ; heavier flywheel – smaller motor but more starting power and hoop stress.

9. Additional Solved Problems

P1 (natural frequency of a beam). Cantilever steel (E = 200 GPa) 0.5 m long, $I = 8\times10^{-9}$ m⁴, end mass 5 kg : $k = 3EI/l^3 = 3(200\times10^9)(8\times10^{-9})/0.125 = 38.4$ kN/m ⇒ $\omega_n = \sqrt{(38400/5)} = 87.6$ rad/s ⇒ $f = 13.9$ Hz.

P2 (torsional). Disc $I = 0.5\ kg{\cdot}m^2$ on steel shaft $d = 20\ mm$, $l = 0.8\ m$, $G = 80\ GPa$: $J = \pi d^4/32 = 1.571\times10^{-8}\ m^4$; $k_t = GJ/l = 1571\ N{\cdot}m/rad \Rightarrow \omega_n = \sqrt{(1571/0.5)} =$ 56.1 rad/s.

P3 (damping ratio). A system has $c = 1.2\ kN{\cdot}s/m$, $k = 100\ kN/m$, $m = 50\ kg$: $c_c = 2\sqrt{(100000\times50)} = 4472 \Rightarrow \zeta = 0.268$ (under-damped). $\omega_n = 44.7\ rad/s$; $\omega_d = 43.1\ rad/s$.

P4 (resonance). A motor of mass 120 kg on springs of total $k = 120\ kN/m$: $\omega_n = 31.6\ rad/s = 302\ rpm$; to avoid resonance operate at 1440 rpm ($r = 4.77$) $\Rightarrow$ TR (undamped) $= 1/(4.77^2 - 1) =$ 0.046 (95 % isolation).

P5 (flywheel speed). $C_s = 0.03$ about $N = 150\ rpm \Rightarrow N_{max} = 150(1 + 0.015) = 152.25$; $N_{min} = 147.75\ rpm$.

Short answers – Quick Revision

- **What is resonance?** Forcing frequency = natural frequency $\Rightarrow$ amplitude limited only by damping ($X = F_0/(2\zeta k)$).
- **Why is isolation poor for $r < \sqrt{2}$?** Transmissibility ≥ 1 – the mount amplifies the force.
- **Why does a single-cylinder engine need a heavy flywheel?** Only the expansion stroke produces positive torque $\Rightarrow$ big ΔE.
- **Difference between flywheel and governor?** Flywheel – cyclic fluctuation (energy reservoir) ; governor – mean speed vs load (control of fuel/steam).
- **What does $\zeta = 1$ mean?** Fastest return to equilibrium without overshoot – critically damped.

10. Quick Formula Sheet – Unit V

Item	Formula
Natural frequency	$\omega_n = \sqrt{(k/m)} = \sqrt{(g/\delta_{st})}$; torsional $\sqrt{(k_t/I)}$
Damping ratio	$\zeta = c/(2\sqrt{(km)})$; $\omega_d = \omega_n\sqrt{(1-\zeta^2)}$
Log decrement	$\delta = 2\pi\zeta/\sqrt{(1-\zeta^2)} \approx 2\pi\zeta$
Forced amplitude	$X = (F_0/k)/\sqrt{((1-r^2)^2+(2\zeta r)^2)}$
Transmissibility	$TR = \sqrt{(1+(2\zeta r)^2)}/\sqrt{((1-r^2)^2+(2\zeta r)^2)}$; isolation if $r > \sqrt{2}$
Mean torque	$T_{mean} = W_{cycle}/(2\pi$ or $4\pi)$

Item	Formula
Flywheel	$\Delta E = I\omega^2 C_s$; $C_s = (\omega_1 - \omega_2)/\omega_{mean}$
Rim stress	$\sigma = \rho v^2$
Punching	$E = \frac{1}{2} F\, t$; $\Delta E = E(1 - t_p/t_c)$

11. Important Questions

- Define DOF and classify vibrations. Derive the natural frequency of a spring–mass system by Newton's and energy methods.
- Derive the equation of motion of a damped SDOF system and discuss under-damped, critically damped and over-damped cases; define logarithmic decrement.
- Derive the steady-state response of a damped forced system ; draw magnification factor & phase curves ; define resonance.
- Define transmissibility ; derive the expression and show that isolation needs $r > \sqrt{2}$; describe isolation methods.
- Draw and explain the turning-moment diagrams of a steam engine, four-stroke I.C. engine and multi-cylinder engine.
- Define coefficient of fluctuation of energy and speed ; derive $\Delta E = I\omega^2 C_s$; explain the design of a flywheel and the flywheel for a punching press.